

Notes on stability theory

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CHAPTER 1

Stable formulas, both 2-valued and real-valued

These notes use two notions of stability, and this chapter introduces both. The first is classical. A formula is stable when unable to order arbitrarily long sequences, and the basic fact about stable formulas is definability of types. We set this up in Section 1.1, where the definability theorem is stated (see Theorem 1.1.6 below), with the proof deferred to Chapter 2. The second notion appears when a formula is regarded as a $\{0, 1\}$ -valued function. In Section 1.2 we pass from such functions to bounded real-valued ones, and stability becomes a property of iterated limits that is closely tied to reflexive Banach spaces.

1.1. The 2-valued case: φ -types and type definability

Throughout this section and Chapters 2 and 3, T is a fixed first-order theory in a language L , and $\varphi(x, y)$ is a fixed L -formula consistent with T , where x and y are finite tuples of variables. Tuples of elements are written $a, b, \dots$ and are always of the appropriate length. We put $\kappa = |L| + \aleph_0$, and for $u \in \{0, 1\}$ we write $\varphi^1 = \varphi$ and $\varphi^0 = \neg\varphi$.

Definition 1.1.1. Let $M \models T$.

- (a) A φ -formula over M is a formula $\varphi(x, b)^u$ with $b \in M$ and $u \in \{0, 1\}$.
- (b) A complete φ -type over M (in the variables x) is a set p of φ -formulas over M such that
 - (i) p is consistent with the elementary diagram of M , and
 - (ii) for each $b \in M$, exactly one of $\varphi(x, b)$ and $\neg\varphi(x, b)$ belongs to p .

The set of complete φ -types over M is denoted $S_\varphi(M)$.

A complete φ -type p is determined by the set $D_p = \{b \in M : \varphi(x, b) \in p\}$, since $p = \{\varphi(x, b) : b \in D_p\} \cup \{\neg\varphi(x, b) : b \in M \setminus D_p\}$. Every φ -type is of the form $\text{tp}_\varphi(c/M) = \{\varphi(x, b)^u : b \in M, N \models \varphi(c, b)^u\}$ for some element c of an elementary extension $N \succeq M$. Conversely, every such set is a complete φ -type.

Lemma 1.1.2 (finite satisfiability). Let $p \in S_\varphi(M)$. For every finite set $\{\varphi(x, b_l)^{u_l} : l < k\} \subseteq p$ there is $a \in M$ with $M \models \bigwedge_{l < k} \varphi(a, b_l)^{u_l}$.

Proof. The $L(M)$ -sentence $\exists x \bigwedge_{l < k} \varphi(x, b_l)^{u_l}$ is consistent with the elementary diagram of M . That diagram is a complete $L(M)$ -theory, so the sentence belongs to the diagram, that is, holds in M . ☺

Definition 1.1.3. Let $p \in S_\varphi(M)$.

- (a) p is *definable* if there is an $L(M)$ -formula $\psi(y)$, a φ -definition of p , such that for all $b \in M$,

$$\varphi(x, b) \in p \iff M \models \psi(b).$$

Equivalently, $D_p = \psi(M)$ is a definable subset of M .

- (b) A *positive Boolean combination of instances of φ over M* is a formula of the form

$$\bigvee_{i < n} \bigwedge_{a \in W_i} \varphi(a, y),$$

where $n < \omega$ and $W_0, \dots, W_{n-1}$ are finite subsets of M . By convention the empty disjunction is $\perp$ and the empty conjunction is $\top$. Every formula built from formulas $\varphi(a, y)$ ($a \in M$) using only $\wedge$ and $\vee$ is equivalent to one of this form. If φ is quantifier-free, so is every positive Boolean combination of instances of φ .

A φ -type p over M is determined by the set D_p , which records, for each parameter b , whether $\varphi(x, b)$ holds of the element that p describes. In general D_p is just a subset of M , with no reason to be describable inside M . When M is infinite there are far more subsets of M than formulas over M , and a definable type is one whose set D_p is captured by a single formula. The motivating example is a realized type. If $c \in M$, then $\text{tp}_\varphi(c/M)$ is defined by the formula $\varphi(c, y)$. So a definable type behaves like the type of an element of M , even when the type is realized only in an elementary extension.

The typical failure of definability comes from orders. In $(\mathbb{Q}, <)$, let $\varphi(x, y)$ be $x < y$ and let p be the type of $\sqrt{2}$ over $\mathbb{Q}$. Then $D_p = \{b \in \mathbb{Q} : b > \sqrt{2}\}$, and p is not definable, since the subsets of $\mathbb{Q}$ definable in $(\mathbb{Q}, <)$ with parameters are finite unions of intervals with rational or infinite endpoints. By contrast, the type of an element just above a rational q is defined by $y > q$.

Definability also has a practical side. A formula $\psi(y)$ makes sense in every elementary extension $N \succeq M$, so a definition of p tells us how to extend p to parameters in N : put $\varphi(x, b)$ in the extension exactly when $N \models \psi(b)$. This is the source of the canonical extensions used to define independence in Chapter 4.

The notion that governs definability is the order property.

Definition 1.1.4. Let $M \models T$ and $n \geq 1$. An n -ladder for φ in M consists of tuples $a_1, \dots, a_n$ and $b_1, \dots, b_n$ of elements of M such that, for all $1 \leq i, j \leq n$,

$$M \models \varphi(a_i, b_j) \iff i \leq j.$$

We say that φ has the *order property* if for every n some model of T contains an n -ladder for φ , and that φ is *stable* otherwise.

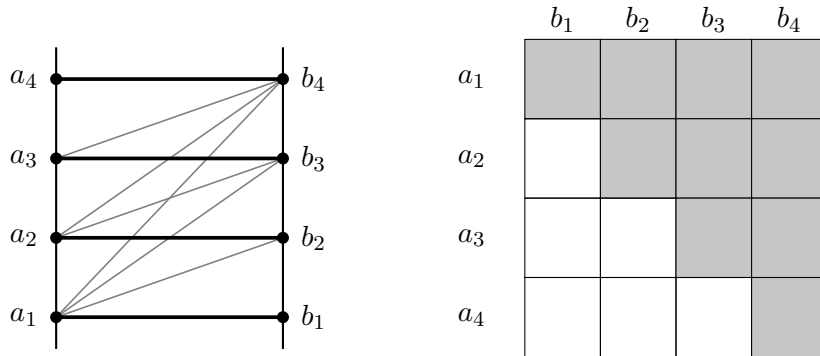


FIGURE 1. A 4-ladder. Left: a_i and b_j are joined when $\varphi(a_i, b_j)$ holds, i.e. when $i \leq j$. The pairs (a_i, b_i) are the rungs (thick), and each a_i also reaches every b_j with $j > i$ (gray). Right: the truth table of $\varphi(a_i, b_j)$, with shaded cells where φ holds. The table is a staircase, which is why an n -ladder is also called a *half-graph*.

Figure 1 explains the name. Put the a_i on one rail and the b_j on the other, both in increasing order, and join a_i to b_j whenever $\varphi(a_i, b_j)$ holds. The “horizontal” pairs (a_i, b_i) form the rungs of a ladder. The defining feature is the asymmetry between the two directions:

a_i is related to every b_j with $j \geq i$ and to none with $j < i$. So φ linearly orders the rungs: $i \leq j$ if and only if $\varphi(a_i, b_j)$.

The order property is a condition on configurations of elements in models of T ; the order property is not a formula, and plays no role in the definitions of types, which are produced in Section 2.2 and are positive Boolean combinations of instances of φ .



Remark 1.1.5. An $(n+1)$ -ladder contains an n -ladder. Hence, by compactness, φ has the order property if and only if some model of T contains an infinite sequence $(a_i, b_i)_{i < \omega}$ with $\varphi(a_i, b_j) \iff i \leq j$. The choice of “ $\leq$ ” is immaterial. Suppose $\varphi(a_i, b_j) \iff i > j$ for $1 \leq i, j \leq n$. Reversing the indices, $a'_i = a_{n+1-i}$ and $b'_j = b_{n+1-j}$, gives $\varphi(a'_i, b'_j) \iff i < j$. Shifting, $b''_j = b'_{j+1}$, then gives $\varphi(a'_i, b''_j) \iff i \leq j$ for $1 \leq i, j \leq n-1$. So a “ $>$ ”-pattern of length n yields an $(n-1)$ -ladder.

Theorem 1.1.6 (Definability of types). *The following are equivalent.*

- (1) φ does not have the order property.
- (2) For every $M \models T$, every $p \in S_\varphi(M)$ has a φ -definition that is a positive Boolean combination of instances $\varphi(a, y)$ with $a \in M$.
- (3) For every $M \models T$, every $p \in S_\varphi(M)$ is definable.

In short: *all φ -types are definable if and only if φ does not have the order property.* The implication (1) $\Rightarrow$ (2) is proved in Section 2.2, and (2) $\Rightarrow$ (3) is trivial. For (3) $\Rightarrow$ (1) we show in Sections 2.3–2.4 that if φ has the order property, then some φ -type is not definable. The route passes through binary trees.

1.2. The real-valued case

In this section we follow Iovino [49]. A formula is identified with a $\{0, 1\}$ -valued function, and stability becomes a property of bounded real-valued functions: the exchangeability of iterated limits. This point of view connects stability with reflexivity of Banach spaces, via a characterization of reflexivity due to James [50] that closely resembles the order property.

1.2.1. Stability and iterated limit exchange. Let M be an L -structure and $\varphi(x, y)$ an L -formula. We identify φ with the function $\varphi: M^{|x|} \times M^{|y|} \rightarrow \{0, 1\}$ that takes the value 1 at (a, b) if and only if $M \models \varphi(a, b)$. More generally, a function $\varphi: A \times B \rightarrow \{0, 1\}$ on arbitrary sets A, B has the *order property* if there are sequences $(a_m)_{m < \omega}$ in A and $(b_n)_{n < \omega}$ in B such that

$$\varphi(a_m, b_n) = 1 \iff m \leq n.$$

Definition 1.2.1. A function $\varphi: A \times B \rightarrow \{0, 1\}$ is *unstable* if either φ or $\neg\varphi = 1 - \varphi$ has the order property, and *stable* otherwise. A formula $\varphi(x, y)$ is *stable on M* if the corresponding function on $M^{|x|} \times M^{|y|}$ is stable.



Remark 1.2.2. A formula $\varphi(x, y)$ is stable in the sense of Definition 1.1.4 if and only if φ is stable on every model of T . Indeed, if φ has the order property, then by compactness (see Remark 1.1.5) some model contains an infinite ladder, so φ is unstable on that model. Conversely, if φ or $\neg\varphi$ has the order property on some $M \models T$, then M contains n -ladders for φ for every n : directly in the first case, and by Remark 1.1.5 in the second, since $\neg\varphi(a_m, b_n) \iff m \leq n$ means $\varphi(a_m, b_n) \iff m > n$.

We now recall convergence along ultrafilters. Let X be a topological space, $(a_i)_{i \in I}$ a family in X , $a \in X$, and $\mathcal{U}$ an ultrafilter on I . We write

$$(1.1) \quad \lim_{i, \mathcal{U}} a_i = a$$

if for every neighborhood O of a we have $\{i \in I : a_i \in O\} \in \mathcal{U}$. If X is Hausdorff, such a limit, if one exists, is unique. If $A \subseteq X$, then $a \in \bar{A}$ if and only if there are a family $(a_i)_{i \in I}$ in A and an ultrafilter $\mathcal{U}$ on I such that (1.1) holds. The space X is compact if and only if $\lim_{i, \mathcal{U}} a_i$ exists for every family $(a_i)_{i \in I}$ in X and every ultrafilter $\mathcal{U}$ on I . In particular, every bounded family of real numbers has a limit along every ultrafilter. Finally, a map $f: X \rightarrow Y$ is continuous if and only if $f(\lim_{i, \mathcal{U}} a_i) = \lim_{i, \mathcal{U}} f(a_i)$ whenever the left-hand limit exists.

The next proposition extracts sequences from iterated limits taken along arbitrary families. Proposition 1.2.3 is used several times below.

Proposition 1.2.3. *Let $\varphi: A \times B \rightarrow [0, 1]$, let $(a_i)_{i \in I}$ be a family in A and $(b_j)_{j \in J}$ a family in B , and let $\mathcal{U}, \mathcal{V}$ be ultrafilters on I, J with*

$$\lim_{i, \mathcal{U}} \lim_{j, \mathcal{V}} \varphi(a_i, b_j) = \alpha, \quad \lim_{j, \mathcal{V}} \lim_{i, \mathcal{U}} \varphi(a_i, b_j) = \beta.$$

Then there are $i_0, i_1, \dots \in I$ and $j_0, j_1, \dots \in J$ such that

$$|\varphi(a_{i_m}, b_{j_n}) - \alpha| \leq 2^{1-m} \text{ if } m < n, \quad |\varphi(a_{i_m}, b_{j_n}) - \beta| \leq 2^{1-n} \text{ if } n < m.$$

Proof. By definition of the limits, for every $\varepsilon > 0$ there are $U_\varepsilon \in \mathcal{U}$ and $V_\varepsilon \in \mathcal{V}$ such that

$$i \in U_\varepsilon \implies \left| \lim_{j, \mathcal{V}} \varphi(a_i, b_j) - \alpha \right| \leq \varepsilon,$$

$$j \in V_\varepsilon \implies \left| \lim_{i, \mathcal{U}} \varphi(a_i, b_j) - \beta \right| \leq \varepsilon.$$

Also, for all $\hat{i} \in I, \hat{j} \in J$ and $\varepsilon > 0$ there are $V_\varepsilon^{\hat{i}} \in \mathcal{V}$ and $U_\varepsilon^{\hat{j}} \in \mathcal{U}$ such that

$$j \in V_\varepsilon^{\hat{i}} \implies \left| \varphi(a_{\hat{i}}, b_j) - \lim_{j, \mathcal{V}} \varphi(a_{\hat{i}}, b_j) \right| \leq \varepsilon,$$

$$i \in U_\varepsilon^{\hat{j}} \implies \left| \varphi(a_i, b_{\hat{j}}) - \lim_{i, \mathcal{U}} \varphi(a_i, b_{\hat{j}}) \right| \leq \varepsilon.$$

Choose recursively

$$i_N \in U_{2^{-N}} \cap \bigcap_{k < N} U_{2^{-k}}^{j_k}, \quad j_N \in V_{2^{-N}} \cap \bigcap_{k \leq N} V_{2^{-k}}^{i_k}.$$

These sets are finite intersections of members of an ultrafilter, hence nonempty. If $m < n$, then $j_n \in V_{2^{-n}}^{i_m}$ and $i_m \in U_{2^{-m}}$, so $|\varphi(a_{i_m}, b_{j_n}) - \alpha| \leq 2^{-n} + 2^{-m} \leq 2^{1-m}$. If $n < m$, then $i_m \in U_{2^{-m}}^{j_n}$ and $j_n \in V_{2^{-n}}$, so $|\varphi(a_{i_m}, b_{j_n}) - \beta| \leq 2^{-m} + 2^{-n} \leq 2^{1-n}$. $\square$

Proposition 1.2.4. *Let $\varphi: A \times B \rightarrow \{0, 1\}$. The following are equivalent.*

- (1) φ is stable.
- (2) For all sequences (a_m) in A and (b_n) in B and all ultrafilters $\mathcal{U}, \mathcal{V}$ on $\mathbb{N}$,

$$\lim_{m, \mathcal{U}} \lim_{n, \mathcal{V}} \varphi(a_m, b_n) = \lim_{n, \mathcal{V}} \lim_{m, \mathcal{U}} \varphi(a_m, b_n).$$

Proof. (2) $\Rightarrow$ (1). Suppose $\varphi(a_m, b_n) = 1 \iff m \leq n$, and let $\mathcal{U} = \mathcal{V}$ be a nonprincipal ultrafilter on $\mathbb{N}$. For fixed m , $\varphi(a_m, b_n) = 1$ for all $n \geq m$, so the left-hand side of (2) is 1. For fixed n , $\varphi(a_m, b_n) = 0$ for all $m > n$, so the right-hand side is 0. If instead $\neg\varphi$ has the order property, the two sides are 0 and 1.

(1) $\Rightarrow$ (2). Suppose the two iterated limits are $\alpha \neq \beta$. Both lie in $\{0, 1\}$. By Proposition 1.2.3 there are sequences (a'_m) in A and (b'_n) in B with $|\varphi(a'_m, b'_n) - \alpha| \leq 2^{1-m}$ for $m < n$ and $|\varphi(a'_m, b'_n) - \beta| \leq 2^{1-n}$ for $n < m$. Since φ is $\{0, 1\}$ -valued, $\varphi(a'_m, b'_n) = \alpha$ for $2 \leq m < n$ and $\varphi(a'_m, b'_n) = \beta$ for $2 \leq n < m$. Put $c_m = a'_{2m+2}$ and $d_n = b'_{2n+3}$. Since

$2m + 2 < 2n + 3 \iff m \leq n$, we get $\varphi(c_m, d_n) = \alpha$ if $m \leq n$ and $\varphi(c_m, d_n) = \beta$ if $m > n$. So φ has the order property if $\alpha = 1$, and $\neg\varphi$ does if $\alpha = 0$. ☺

Proposition 1.2.4 motivates the following definition.

Definition 1.2.5. A bounded function $\varphi: A \times B \rightarrow \mathbb{R}$ is *stable* if, for all sequences (a_m) in A and (b_n) in B and all ultrafilters $\mathcal{U}, \mathcal{V}$ on $\mathbb{N}$,

$$\lim_{m, \mathcal{U}} \lim_{n, \mathcal{V}} \varphi(a_m, b_n) = \lim_{n, \mathcal{V}} \lim_{m, \mathcal{U}} \varphi(a_m, b_n).$$

By Proposition 1.2.4, for $\{0, 1\}$ -valued functions this agrees with Definition 1.2.1. By Proposition 1.2.3, the sequences in Definition 1.2.5 can be replaced by arbitrary families. For a bounded set of continuous functions on a compact space, the same condition is the hypothesis of Grothendieck's double limit criterion (see Appendix A below).

Corollary 1.2.6. *If $\varphi: A \times B \rightarrow [0, 1]$ is stable, then $\lim_{i, \mathcal{U}} \lim_{j, \mathcal{V}} \varphi(a_i, b_j) = \lim_{j, \mathcal{V}} \lim_{i, \mathcal{U}} \varphi(a_i, b_j)$ for all families $(a_i)_{i \in I}$, $(b_j)_{j \in J}$ and all ultrafilters $\mathcal{U}$ on I , $\mathcal{V}$ on J .*

Proof. If the two limits are $\alpha \neq \beta$, take (i_m) , (j_n) as in Proposition 1.2.3 and a nonprincipal ultrafilter $\mathcal{W}$ on $\mathbb{N}$. For fixed m , $\lim_{n, \mathcal{W}} \varphi(a_{i_m}, b_{j_n})$ is within 2^{1-m} of α , so $\lim_{m, \mathcal{W}} \lim_{n, \mathcal{W}} \varphi(a_{i_m}, b_{j_n}) = \alpha$. Symmetrically, the other iterated limit is β , which contradicts the stability of φ . ☺

1.2.2. Stability and continuous extendability.

Definition 1.2.7. Let $\varphi: A \times B \rightarrow [0, 1]$. For $a \in A$, the *left φ -type* of a is the function $\text{ltp}_\varphi(a): y \mapsto \varphi(a, y)$ on B . For $b \in B$, the *right φ -type* of b is the function $\text{rtp}_\varphi(b): x \mapsto \varphi(x, b)$ on A . The *space of left φ -types* $\text{LS}(\varphi)$ is the closure of $\{\text{ltp}_\varphi(a) : a \in A\}$ in $[0, 1]^B$ with the product topology, and the *space of right φ -types* $\text{RS}(\varphi)$ is the closure of $\{\text{rtp}_\varphi(b) : b \in B\}$ in $[0, 1]^A$. Both spaces are compact.

Proposition 1.2.8. *For $\varphi: A \times B \rightarrow [0, 1]$ the following are equivalent.*

- (1) φ is stable.
- (2) There is a separately continuous function $\hat{\varphi}: \text{LS}(\varphi) \times \text{RS}(\varphi) \rightarrow [0, 1]$ such that

$$\hat{\varphi}(\text{ltp}_\varphi(a), \text{rtp}_\varphi(b)) = \varphi(a, b) \quad \text{for all } (a, b) \in A \times B.$$

Proof. (1) $\Rightarrow$ (2). Let $p \in \text{LS}(\varphi)$ and $q \in \text{RS}(\varphi)$. There are families $(a_i)_{i \in I}$ in A , $(b_j)_{j \in J}$ in B and ultrafilters $\mathcal{U}$ on I , $\mathcal{V}$ on J with

$$p = \lim_{i, \mathcal{U}} \text{ltp}_\varphi(a_i), \quad q = \lim_{j, \mathcal{V}} \text{rtp}_\varphi(b_j).$$

By Corollary 1.2.6, $\lim_{i, \mathcal{U}} \lim_{j, \mathcal{V}} \varphi(a_i, b_j) = \lim_{j, \mathcal{V}} \lim_{i, \mathcal{U}} \varphi(a_i, b_j)$. Define $\hat{\varphi}(p, q)$ to be this common value. Since limits in the product topology are computed coordinatewise, this common value satisfies

$$(1.2) \quad \hat{\varphi}(p, q) = \lim_{i, \mathcal{U}} q(a_i) = \lim_{j, \mathcal{V}} p(b_j).$$

$\hat{\varphi}$ is well defined. Suppose also $p = \lim_{i, \mathcal{U}'} \text{ltp}_\varphi(a'_i)$ and $q = \lim_{j, \mathcal{V}'} \text{rtp}_\varphi(b'_j)$. Then $\lim_{j, \mathcal{V}} \varphi(x, b_j) = q(x) = \lim_{j, \mathcal{V}'} \varphi(x, b'_j)$ for every $x \in A$, and $\lim_{i, \mathcal{U}} \varphi(a_i, y) = p(y) = \lim_{i, \mathcal{U}'} \varphi(a'_i, y)$ for every $y \in B$. Hence, using Corollary 1.2.6 for the middle step,

$$\lim_{i, \mathcal{U}} \lim_{j, \mathcal{V}} \varphi(a_i, b_j) = \lim_{i, \mathcal{U}} \lim_{j, \mathcal{V}'} \varphi(a_i, b'_j) = \lim_{j, \mathcal{V}'} \lim_{i, \mathcal{U}} \varphi(a_i, b'_j) = \lim_{j, \mathcal{V}'} \lim_{i, \mathcal{U}'} \varphi(a'_i, b'_j).$$

$\hat{\varphi}$ extends φ . Take constant families.

$\hat{\varphi}$ is separately continuous. Fix p , and let $q = \lim_{k, \mathcal{W}} q_k$ in $\text{RS}(\varphi)$, where $\mathcal{W}$ is an ultrafilter on a set K . Write $q_k = \lim_{j, \mathcal{V}_k} \text{rtp}_\varphi(b_{k,j})$ with j ranging over a set J_k . On $J' = \{(k, j) : k \in K, j \in J_k\}$ let $\mathcal{V}'$ be the ultrafilter of all $X \subseteq J'$ with $\{k : \{j : (k, j) \in X\} \in \mathcal{V}_k\} \in \mathcal{W}$. For

every family $(z_{k,j})$ in a compact Hausdorff space, $\lim_{(k,j), \mathcal{V}'} z_{k,j} = \lim_{k, \mathcal{W}} \lim_{j, \mathcal{V}_k} z_{k,j}$. Hence $q = \lim_{(k,j), \mathcal{V}'} \text{rtp}_\varphi(b_{k,j})$, and by (1.2),

$$\hat{\varphi}(p, q) = \lim_{(k,j), \mathcal{V}'} p(b_{k,j}) = \lim_{k, \mathcal{W}} \lim_{j, \mathcal{V}_k} p(b_{k,j}) = \lim_{k, \mathcal{W}} \hat{\varphi}(p, q_k).$$

So $\hat{\varphi}(p, \cdot)$ is continuous. Continuity of $\hat{\varphi}(\cdot, q)$ is proved symmetrically.

(2) $\Rightarrow$ (1). Let $(a_m), (b_n)$ be sequences in A, B , let $\mathcal{U}, \mathcal{V}$ be ultrafilters on $\mathbb{N}$, and put $p = \lim_{m, \mathcal{U}} \text{ltp}_\varphi(a_m)$ and $q = \lim_{n, \mathcal{V}} \text{rtp}_\varphi(b_n)$. Then $(p, q) \in \text{LS}(\varphi) \times \text{RS}(\varphi)$, and by separate continuity

$$\lim_{m, \mathcal{U}} \lim_{n, \mathcal{V}} \varphi(a_m, b_n) = \lim_{m, \mathcal{U}} \hat{\varphi}(\text{ltp}_\varphi(a_m), q) = \hat{\varphi}(p, q) = \lim_{n, \mathcal{V}} \lim_{m, \mathcal{U}} \varphi(a_m, b_n). \quad \text{☺}$$

1.2.3. Stability and reflexivity. For a Banach space E with dual E^* , we write B_E for the closed unit ball of E and $\langle x, y \rangle$ for $y(x)$, where $x \in E$ and $y \in E^*$. Recall that E is *reflexive* if the canonical embedding of E into E^{**} is surjective.

Theorem 1.2.9 (Iovino). *For $\varphi: A \times B \rightarrow [0, 1]$ the following are equivalent.*

- (1) φ is stable.
- (2) There are a reflexive Banach space E and maps $u: A \rightarrow B_E$ and $v: B \rightarrow B_{E^*}$ such that

$$\varphi(x, y) = \langle u(x), v(y) \rangle \quad \text{for all } (x, y) \in A \times B.$$

Proof. See [49, Theorem 3.3]. For (1) $\Rightarrow$ (2), stability is used to show that the operator $T: \ell^\infty(A)^* \rightarrow \ell^\infty(B)$ given by $T\mu(y) = \int \varphi(x, y) d\mu(x)$ is weakly compact; this step goes back to Krivine and Maurey [51] and Raynaud [52]. The theorem of Davis, Figiel, Johnson and Pełczyński [46] then factors T through a reflexive space. The implication (2) $\Rightarrow$ (1) follows from the weak compactness of B_E and the weak* compactness of B_{E^*} . ☺

By Remark 1.2.2 and Proposition 1.2.4, a formula $\varphi(x, y)$ is stable in the sense of Definition 1.1.4 if and only if, in every model $M \models T$, the function φ on M is the restriction of the pairing map $B_E \times B_{E^*} \rightarrow \mathbb{R}$ for some reflexive Banach space E (depending on M), composed with maps $M^{|x|} \rightarrow B_E$ and $M^{|y|} \rightarrow B_{E^*}$.

1.2.4. Stability and polynomial approximation. Theorem 1.1.6 says that a φ -type over a model of a stable theory is defined by a Boolean combination of instances $\varphi(a, y)$. On $\{0, 1\}$ the Boolean connectives are polynomials: $\neg t = 1 - t$ and $t \wedge s = ts$. So the real-valued analogue of a definition is a *uniform limit of polynomials in instances of φ* . The Stone–Weierstrass theorem shows that stability is exactly what is needed for such approximations to exist. This is the approach to definability taken in continuous logic, where Stone–Weierstrass shows that the continuous functions on a type space are exactly the uniform limits of formulas [43, Propositions 1.4, 3.4 and 3.10]. Polynomial approximation is also the approach of Ben Yaacov [42], who derives the definability of types in stable theories from Grothendieck’s double-limit criterion [44] (see Appendix A below). Grothendieck’s criterion is the same interchange of iterated limits as Definition 1.2.5.

Fact 1.2.10 (Stone–Weierstrass). *Let K be a compact Hausdorff space and let $\mathcal{F} \subseteq C(K)$ be a family of continuous real-valued functions that separates the points of K . Then every $f \in C(K)$ is a uniform limit on K of functions of the form $P(g_1, \dots, g_k)$, where $g_1, \dots, g_k \in \mathcal{F}$ and P is a real polynomial in k variables.*

Fix $\varphi: A \times B \rightarrow [0, 1]$. For $a \in A$, let $e_a: \text{RS}(\varphi) \rightarrow [0, 1]$ be the evaluation map $e_a(q) = q(a)$, and for $b \in B$, let $e_b: \text{LS}(\varphi) \rightarrow [0, 1]$ be $e_b(p) = p(b)$. These maps are continuous, because $\text{RS}(\varphi)$ and $\text{LS}(\varphi)$ carry the product topology. On realized types, these maps are

instances of φ : $e_a(\text{rtp}_\varphi(b)) = \varphi(a, b) = e_b(\text{lt}_\varphi(a))$. The family $\{e_a : a \in A\}$ separates the points of $\text{RS}(\varphi)$, since two distinct functions $q \neq q'$ in $[0, 1]^A$ differ at some $a \in A$. Likewise $\{e_b : b \in B\}$ separates the points of $\text{LS}(\varphi)$.

Theorem 1.2.11. *For $\varphi : A \times B \rightarrow [0, 1]$ the following are equivalent.*

- (1) φ is stable.
- (2) *There is a function $\hat{\varphi} : \text{LS}(\varphi) \times \text{RS}(\varphi) \rightarrow [0, 1]$ with $\hat{\varphi}(\text{lt}_\varphi(a), \text{rtp}_\varphi(b)) = \varphi(a, b)$ for all $(a, b) \in A \times B$, such that for every $\varepsilon > 0$:*
 - (a) *for every $p \in \text{LS}(\varphi)$ there are $a_1, \dots, a_k \in A$ and a real polynomial P with*

$$|\hat{\varphi}(p, q) - P(q(a_1), \dots, q(a_k))| \leq \varepsilon \quad \text{for all } q \in \text{RS}(\varphi);$$

- (b) *for every $q \in \text{RS}(\varphi)$ there are $b_1, \dots, b_l \in B$ and a real polynomial Q with*

$$|\hat{\varphi}(p, q) - Q(p(b_1), \dots, p(b_l))| \leq \varepsilon \quad \text{for all } p \in \text{LS}(\varphi).$$

In this case, $\hat{\varphi}$ is the function of Proposition 1.2.8. Taking $q = \text{rtp}_\varphi(b)$ in (a) gives

$$|p(b) - P(\varphi(a_1, b), \dots, \varphi(a_k, b))| \leq \varepsilon \quad \text{for all } b \in B.$$

So every left φ -type is a uniform limit on B of polynomials in instances $\varphi(a, y)$ with $a \in A$, and symmetrically for right types.

Proof. (1) $\Rightarrow$ (2). Let $\hat{\varphi}$ be the separately continuous function given by Proposition 1.2.8. For fixed p , the function $\hat{\varphi}(p, \cdot)$ is continuous on the compact Hausdorff space $\text{RS}(\varphi)$. Applying Fact 1.2.10 with $\mathcal{F} = \{e_a : a \in A\}$ gives (a). Clause (b) follows in the same way, using the continuity of $\hat{\varphi}(\cdot, q)$ on $\text{LS}(\varphi)$ and $\mathcal{F} = \{e_b : b \in B\}$. For the displayed consequence, note that $p(b) = \hat{\varphi}(p, \text{rtp}_\varphi(b))$. Indeed, if $p = \lim_{i, \mathcal{U}} \text{lt}_\varphi(a_i)$, then continuity in the first variable gives $\hat{\varphi}(p, \text{rtp}_\varphi(b)) = \lim_{i, \mathcal{U}} \varphi(a_i, b) = p(b)$.

(2) $\Rightarrow$ (1). Each function $q \mapsto P(q(a_1), \dots, q(a_k))$ is continuous on $\text{RS}(\varphi)$. By (a), $\hat{\varphi}(p, \cdot)$ is a uniform limit of such functions, hence continuous. By (b), $\hat{\varphi}(\cdot, q)$ is continuous for the same reason. Hence $\hat{\varphi}$ is separately continuous, and φ is stable by Proposition 1.2.8. ☺

The approximation in Theorem 1.2.11 is required on all of $\text{RS}(\varphi)$ (respectively $\text{LS}(\varphi)$), not only at realized types. The next example shows that approximation on B alone does not imply stability.

Example 1.2.12. Let $A = B = \mathbb{N}$ and $\varphi(m, n) = 1$ if $m \leq n$, $\varphi(m, n) = 0$ otherwise. Then φ has the order property and is therefore unstable. Here $\text{lt}_\varphi(m)$ is the indicator function of $[m, \infty)$, and $\text{RS}(\varphi)$ consists of the indicator functions of the intervals $[0, n]$ together with the constant function $\mathbf{1}$. Similarly, $\text{LS}(\varphi)$ consists of the functions $\text{lt}_\varphi(m)$ together with the constant function $\mathbf{0}$. So every $p \in \text{LS}(\varphi)$, viewed as a function on B , is either an instance $\varphi(m, \cdot)$ or the constant 0, which is trivially a polynomial in instances. Symmetrically, every $q \in \text{RS}(\varphi)$ is an instance or the constant 1. The failure appears at the unrealized pair $(\mathbf{0}, \mathbf{1})$. If $\hat{\varphi}$ were separately continuous and extended φ as in Theorem 1.2.11, then $\hat{\varphi}(\mathbf{0}, \text{rtp}_\varphi(n)) = \mathbf{0}(n) = 0$ for all n , and $\text{rtp}_\varphi(n) \rightarrow \mathbf{1}$, so $\hat{\varphi}(\mathbf{0}, \mathbf{1}) = 0$. But also $\hat{\varphi}(\text{lt}_\varphi(m), \mathbf{1}) = \mathbf{1}(m) = 1$ for all m , and $\text{lt}_\varphi(m) \rightarrow \mathbf{0}$, so $\hat{\varphi}(\mathbf{0}, \mathbf{1}) = 1$.

Corollary 1.2.13. *Let M be an L -structure and let $\varphi(x, y)$ be a formula that is stable on M . Then every $p \in \text{LS}(\varphi)$ is defined by a Boolean combination of instances: there are $a_1, \dots, a_k \in M^{|x|}$ and a Boolean combination $\psi(y)$ of $\varphi(a_1, y), \dots, \varphi(a_k, y)$ such that, for all $b \in M^{|y|}$,*

$$p(b) = 1 \iff M \models \psi(b).$$

If $M \models T$, then $\text{LS}(\varphi)$ is the set of φ -types p over M , each identified with the characteristic function of the set D_p on $M^{|y|}$. Every such type is finitely satisfiable in M , hence a pointwise limit of realized types. So every φ -type over a model is definable.

Proof. Apply Theorem 1.2.11 with $\varepsilon = \frac{1}{3}$ to get $a_1, \dots, a_k$ and P . Both $p(b)$ and $\varphi(a_i, b)$ lie in $\{0, 1\}$, so $p(b) = 1$ if and only if $t = (\varphi(a_1, b), \dots, \varphi(a_k, b))$ belongs to $S = \{t \in \{0, 1\}^k : P(t) > \frac{1}{2}\}$. Take $\psi(y) = \bigvee_{t \in S} \bigwedge_{i \leq k} \varphi(a_i, y)^{t_i}$, with the notation $\varphi^1 = \varphi$ and $\varphi^0 = \neg\varphi$. ☺

By Remark 1.2.2, a formula without the order property is stable on every model of T . So Corollary 1.2.13 gives (1) $\Rightarrow$ (3) of Theorem 1.1.6. The definition produced by Corollary 1.2.13 is a Boolean combination of instances rather than a positive one, of a size depending on p .



Remark 1.2.14. (1) *Lattice version.* The lattice form of the Stone–Weierstrass theorem replaces polynomials by the operations max, min and affine maps. This is the form used in continuous logic, where the lattice form shows that a small set of connectives generates all continuous connectives up to uniform approximation [43, Proposition 1.4].

(2) *Convex combinations.* Stability gives more than a polynomial approximation. By Grothendieck’s criterion (see Proposition A.1.6 below), stability of φ implies that $\{e_a : a \in A\}$ is relatively weakly compact in $C(\text{RS}(\varphi))$. In that case $p = \lim_{i, \mathcal{U}} \text{ltp}_\varphi(a_i)$ implies that $\hat{\varphi}(p, \cdot)$ is the weak limit of the functions e_{a_i} along $\mathcal{U}$. Mazur’s lemma then shows that $\hat{\varphi}(p, \cdot)$ is a uniform limit of convex combinations $\sum_j \lambda_j e_{a_{i_j}}$. So p is a uniform limit on B of averages of instances $\varphi(a, y)$. Ben Yaacov [42] uses this convex approximation to recover the familiar forms of the defining formulas.

1.3. Historical remarks

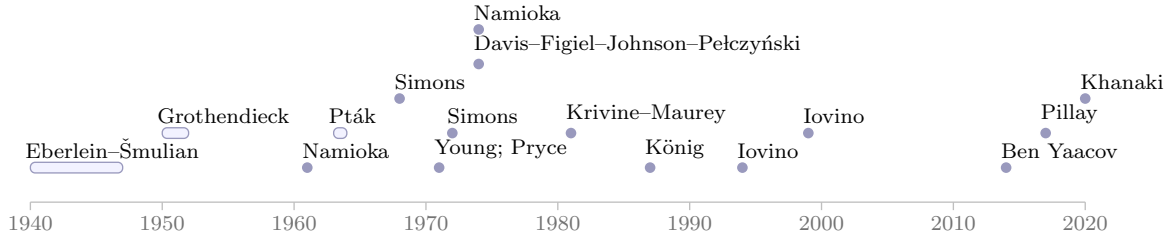


Section 1.1. Counting types begins with Morley’s proof of Łoś’s conjecture [35]. There he introduced totally transcendental theories, defined by a restriction on the topology of the type spaces of those theories, and showed that for countable theories total transcendence is equivalent to a restriction on the number of types: only countably many types over countable sets. This is the condition now called ω -stability. The general notion arose soon after. Rowbottom set out to extend Morley’s categoricity theorem to uncountable theories, and in the process he generalized ω -stability to λ -stability: a theory is λ -stable if there are at most λ types over sets of size λ [66]. Shelah then called a theory *stable* when λ -stable for some λ , and began the study of stable theories in general in [41]. In [36] he moved from theories to single formulas. There he defined when a formula $\varphi(x; y)$ is *stable in λ* : φ has at most λ φ -types over sets of size λ [36, Definition 2.1]. He also defined the *order property* [36, Definition 2.2] and proved that stability is equivalent to the absence of the order property. The ladder formulation of Definition 1.1.4 is a finitary restatement of Shelah’s condition. Definable types were introduced by Gaifman in his study of end extensions of models of Peano arithmetic [40], and Shelah arrived at them independently. The theorem that φ -types are definable when φ is stable is [36, Theorem 3.1]; see also [57, Chapter II]. Today this theorem is often called the fundamental theorem of stability theory. Ben Yaacov [42] observed that the same result follows from Grothendieck’s 1952 work on weak compactness in function spaces [44] (see Appendix A below). This is the viewpoint of Section 1.2.



Section 1.2.1. Interchanging iterated limits is an old theme. For double sequences of reals, the classical sufficient condition is uniform convergence in one variable (the Moore–Osgood theorem). The condition of Definition 1.2.5 requires the iterated limits to agree along *all*

pairs of sequences. This condition comes from functional analysis, and reached model theory through Banach space geometry.



1940–1947

Eberlein–Šmulian. Šmulian (1940) and Eberlein [17] (1947) prove that in a Banach space weak compactness and weak sequential compactness coincide, so that weak compactness can be tested along sequences.

1950–1952

Grothendieck. Grothendieck proves the *double limit criterion*, announced in [18] and proved in [44]. For compact K , a bounded set $H \subseteq C(K)$ is relatively weakly compact if and only if $\lim_m \lim_n f_n(x_m) = \lim_n \lim_m f_n(x_m)$ for all sequences (f_n) in H and (x_m) in K for which both iterated limits exist. For an arbitrary topological space, the same condition decides when pointwise limits of functions in H are continuous. See Appendix A.

1961

Namioka. Namioka’s “substitute for Lebesgue’s bounded convergence theorem” [23], a combinatorial result from which Krein’s theorem on weak compactness of closed convex hulls follows without measure theory.

1963–1964

Pták. Pták [26] finds a purely combinatorial lemma on the existence of convex means, from which the double limit criterion and Krein’s theorem follow. In [27] he characterizes, again by an interchange of iterated limits, the bounded separately continuous functions on a product $X \times Y$ that extend to separately weak*-continuous bilinear forms.

1968

Simons. Simons [28] introduces the *Fubini characterization* of the interchange of double limits. A bounded function on a product satisfies the iterated limit condition if and only if the two iterated integrals of the function agree, for every pair of measures on the factors. This characterization plays an important role in the proof of the factorization theorem of Davis, Figiel, Johnson and Pełczyński [46].

1971

Young; Pryce. N. J. Young [32] gives topological proofs of Pták’s double-limit theorems, based on Grothendieck’s work rather than combinatorics, and generalizes Pták’s bilinear extension theorem. Pryce [25] applies a device of Whitley to pointwise compactness in spaces of continuous functions.

1972

Simons. In three consecutive papers, Simons gives an elementary proof of Pták’s lemma [31]. He also proves the “sup-limsup” theorem now known as *Simons’ inequality* [29], and relates minimax theorems to James’s characterization of reflexivity [30].

- 1974** • **Davis–Figiel–Johnson–Pełczyński.** Davis, Figiel, Johnson and Pełczyński [46] prove that every weakly compact operator factors through a reflexive Banach space. This is the key step in the proof of Theorem 1.2.9.
- 1974** • **Namioka.** Namioka [24] studies when separately continuous functions on products are jointly continuous at many points. This is the question behind Proposition 1.2.8.
- 1981** • **Krivine–Maurey.** Krivine and Maurey [51] apply the interchange of limits to the norm and define *stable Banach spaces*. They prove that every stable Banach space contains some ℓ_p almost isometrically, generalizing Aldous’s theorem on subspaces of L_1 .
- 1987** • **König.** König [22] studies the double limit relation in angelic spaces, where compactness can be tested by sequences.
- 1994** • **Iovino.** Iovino’s Ph.D. thesis [47] introduces stability for real-valued logic.
- 1999** • **Iovino.** Iovino [49] observes that Shelah’s order property is the 2-valued case of the same condition, and proves Theorem 1.2.9. Propositions 1.2.4 and 1.2.3 are taken from his paper.
- 2014** • **Ben Yaacov.** Ben Yaacov [42] derives the definability of types in stable theories directly from Grothendieck’s criterion.
- 2017** • **Pillay.** Pillay [21] comments on the model-theoretic reading of Grothendieck’s criterion, and derives from the criterion the positive form of the definitions.
- 2020** • **Khanaki.** Khanaki [20] studies stability, NIP and NSOP systematically through topological properties of function spaces.

⌘ **Section 1.2.2.** Spaces of left and right φ -types (see Definition 1.2.7) are the analytic counterpart of Shelah’s spaces of φ -types. The use of *types* in Banach space theory, as limits of functions $y \mapsto \|x_n + y\|$, is due to Krivine and Maurey [51]. Proposition 1.2.8, which characterizes stability by the existence of a separately continuous extension to pairs of types, is Proposition 2.6 of Iovino [49]. Separately continuous functions on products of compact spaces, and their relation to iterated limits, are the subject of Grothendieck’s paper [44].

⌘ **Section 1.2.3.** James [50] characterized reflexivity by the absence of sequences (a_m) in B_E and (f_n) in B_{E^*} whose pairing takes one value for $m < n$ and another for $m \geq n$. This is a Banach-space form of the order property. In the early 1980s, Krivine and Maurey [51] and Raynaud [52] showed that stable functions give rise to weakly compact operators. The factorization theorem of Davis, Figiel, Johnson and Pełczyński [46] (1974) says that every weakly compact operator factors through a reflexive space. Iovino [49] combined these results to prove Theorem 1.2.9: stable formulas are exactly the pairings between the unit balls of a reflexive space E and of E^* .

⌘ **Section 1.2.4.** Weierstrass proved in 1885 that every continuous function on a compact interval is a uniform limit of polynomials [39]. Stone replaced the interval by an arbitrary compact Hausdorff space and the coordinate function by any family of functions that separates points [37, 38]. Ben Yaacov and Usvyatsov [43] used Stone’s theorem to lay the foundations of continuous first-order logic. There Stone’s theorem shows that finitely many connectives suffice and that continuous functions on type spaces are exactly the definable predicates. Ben

Yaacov [42] combined Grothendieck's criterion with Mazur's lemma [34] (a weakly convergent sequence has convex combinations converging in norm). From this he derived the defining formulas of stable types as uniform limits of averages of instances; see Remark 1.2.14.

CHAPTER 2

Stability, trees, and definability of types

In Chapter 1 we stated the theorem on definability of types (see Theorem 1.1.6) without proof. This chapter supplies the proof. The proof passes through binary trees, and along the way shows that a formula is stable exactly when unable to encode tall trees. As in Section 1.1, T is a fixed first-order theory in a language L , and $\varphi(x, y)$ is a fixed L -formula consistent with T .

2.1. The equivalence proved in this chapter

We begin by stating precisely what will be proved. Four conditions on φ turn out to coincide: instability, the order property, the presence of tall binary trees, and the existence of a non-definable φ -type. The notions of φ -type, definable type, ladder, order property and stability were discussed in Section 1.1. The remaining notion, that of a binary tree for φ , is defined below.

Notation 2.1.1. For $u \in \{0, 1\}$ we write $\varphi^1 = \varphi$ and $\varphi^0 = \neg\varphi$. For an ordinal μ , $2^{<\mu}$ is the set of 0–1 sequences of length $<\mu$, and $2^{\leq n} = 2^{<n} \cup 2^n$. We write $\sigma \sqsubseteq \tau$ if σ is an initial segment of τ , $\sigma \sqsubset \tau$ if moreover $\sigma \neq \tau$, $|\sigma|$ for the length of σ , $\sigma^\frown \langle u \rangle$ for the extension of σ by one entry u , and $\langle u \rangle \frown \sigma$ for the sequence obtained by putting u in front of σ . For η of length $>k$, $\eta|k$ is the initial segment of length k .

Definition 2.1.2. A *tree of height n for φ in a model $M \models T$* consists of families $(b_\sigma : \sigma \in 2^{<n})$ and $(a_\eta : \eta \in 2^n)$ of tuples of M such that

$$\sigma^\frown \langle u \rangle \sqsubseteq \eta \implies M \models \varphi(a_\eta, b_\sigma)^u.$$

In other words, the leaf a_η satisfies $\varphi(x, b_\sigma)$ exactly when the branch η turns right ($u = 1$) at the node σ . We say that φ *encodes binary trees of arbitrarily large finite height* if for every n some model of T contains a tree of height n for φ . A tree of height $n + 1$ restricts to one of height n .

Theorem 2.1.3. *Let $\varphi(x, y)$ be an L -formula. The following are equivalent.*

- (1) φ is unstable (see Definition 1.1.4).
- (2) φ has the order property (see Definition 1.1.4).
- (3) φ encodes binary trees of arbitrarily large finite height.
- (4) Some φ -type over some model of T is not definable (see Definitions 1.1.1 and 1.1.3).

The equivalence (1) $\Leftrightarrow$ (2) holds by Definition 1.1.4. The implication (2) $\Rightarrow$ (3) is proved in Section 2.3, as Proposition 2.3.1. The implication (3) $\Rightarrow$ (4) is proved in Section 2.4, as Proposition 2.4.2. The implication (4) $\Rightarrow$ (2) is proved in contrapositive form in Section 2.2, by Theorem 2.2.1 and Corollary 2.2.2.

The next chapter gives alternative proofs of two of these implications, and each adds something the proofs in this chapter do not provide. In Section 3.1, Proposition 3.1.5 and Corollary 3.1.6 prove (3) $\Rightarrow$ (2) directly, by finite combinatorics based on Ramsey's theorem. The proof in this chapter reaches the order property only after passing through non-definable types, which requires counting types over models of large cardinality. The direct proof stays

inside a single model and is quantitative: a tree of height $\binom{2n+2}{n+1} - 1$ in a model M already yields an n -ladder in M . In Section 3.2, Proposition 3.2.3 proves (4) $\Rightarrow$ (3), again in contrapositive form, by the classical rank argument. Proposition 3.2.3 shows that the absence of tall trees alone makes every φ -type definable, without going through the order property, and gives the definition explicitly in terms of a rank. Unlike the definitions of Section 2.2, however, these definitions are not positive Boolean combinations of instances of φ .

2.2. From the absence of the order property to type definability

In this section we prove (1) $\Rightarrow$ (2) of Theorem 1.1.6. The definition we obtain is built from instances of φ itself using only $\wedge$ and $\vee$. In particular, if φ is quantifier-free, so is the definition.

Theorem 2.2.1. *Suppose $\varphi(x, y)$ does not have the order property, and let $M \models T$ and $p \in S_\varphi(M)$. Then there is a finite set $F \subseteq M$ such that, with*

$$D = \{b \in M : \varphi(x, b) \in p\} \quad \text{and} \quad \mathcal{W} = \{\{a \in F : M \models \varphi(a, b)\} : b \in D\},$$

the formula

$$d_p\varphi(y) := \bigvee_{W \in \mathcal{W}} \bigwedge_{a \in W} \varphi(a, y)$$

is a φ -definition of p . In other words, for all $b \in M$, $\varphi(x, b) \in p$ if and only if $M \models d_p\varphi(b)$.

Note that $\mathcal{W}$ is a family of subsets of the finite set F , so $\mathcal{W}$ is finite and $d_p\varphi$ is a genuine formula.

Proof. For $b \in M$ write $S_b = \{a \in M : M \models \varphi(a, b)\}$. The only property of p we use is that p is finitely satisfiable in M (see Lemma 1.1.2).

Claim. There is a finite $F \subseteq M$ such that $S_b \cap F \not\subseteq S_{b'}$ for all $b \in D$ and $b' \in M \setminus D$.

Granting the claim, we check that $d_p\varphi$ defines p . Let $b \in M$.

- (a) If $b \in D$, then $W := S_b \cap F \in \mathcal{W}$ and $W \subseteq S_b$, so $M \models \bigwedge_{a \in W} \varphi(a, b)$. Hence $M \models d_p\varphi(b)$.
- (b) If $b \notin D$ and $W \in \mathcal{W}$, say $W = S_{b_0} \cap F$ with $b_0 \in D$, then $W \not\subseteq S_b$ by the claim. So some $a \in W$ has $M \models \neg\varphi(a, b)$, and $M \models \neg\bigwedge_{a \in W} \varphi(a, b)$. As this holds for every $W \in \mathcal{W}$, $M \models \neg d_p\varphi(b)$.

Proof of the claim. Suppose the claim fails. Then for every finite $F \subseteq M$ there are $b \in D$ and $b' \in M \setminus D$ with $S_b \cap F \subseteq S_{b'}$, i.e. $M \models \varphi(a, b) \rightarrow \varphi(a, b')$ for all $a \in F$. We choose recursively, for $i < \omega$, elements $b_i \in D$, $b'_i \in M \setminus D$ and $a_i \in M$ as follows.

- (a) Apply the failure of the claim to $F = \{a_j : j < i\}$ to get $b_i \in D$ and $b'_i \in M \setminus D$ with

$$(2.1) \quad M \models \varphi(a_j, b_i) \rightarrow \varphi(a_j, b'_i) \quad \text{for all } j < i.$$

- (b) Since $\varphi(x, b_k) \in p$ and $\neg\varphi(x, b'_k) \in p$ for $k \leq i$, Lemma 1.1.2 gives $a_i \in M$ with

$$(2.2) \quad M \models \varphi(a_i, b_k) \wedge \neg\varphi(a_i, b'_k) \quad \text{for all } k \leq i.$$

Color each pair $j < i$ by the pair of truth values $(\varphi(a_j, b_i), \varphi(a_j, b'_i)) \in \{0, 1\}^2$. By (2.1), the color (1, 0) never occurs. By Ramsey's theorem there is an infinite $H = \{h_0 < h_1 < \dots\} \subseteq \omega$ on which all pairs have the same color c . Let $s, t < \omega$.

Case $c \in \{(0, 1), (1, 1)\}$. If $s < t$, then $M \models \varphi(a_{h_s}, b'_{h_t})$ by homogeneity. If $s \geq t$, then $M \models \neg\varphi(a_{h_s}, b'_{h_t})$ by (2.2). So $\varphi(a_{h_s}, b'_{h_t})$ holds if and only if $s < t$, and with $\beta_t = b'_{h_{t+1}}$ we get $M \models \varphi(a_{h_s}, \beta_t) \iff s \leq t$.

Case $c = (0, 0)$. If $s < t$, then $M \models \neg\varphi(a_{h_s}, b_{h_t})$ by homogeneity. If $s \geq t$, then $M \models \varphi(a_{h_s}, b_{h_t})$ by (2.2). So $\varphi(a_{h_s}, b_{h_t})$ holds if and only if $s \geq t$. For a given n , reversing the first n indices, $\alpha_s = a_{h_{n-1-s}}$ and $\beta_t = b_{h_{n-1-t}}$, gives $M \models \varphi(\alpha_s, \beta_t) \iff s \leq t$ for $s, t < n$.

In both cases M contains an n -ladder for every n , so φ has the order property, a contradiction. This proves the claim and the theorem. ☺

Corollary 2.2.2. *If φ does not have the order property, then every $p \in S_\varphi(M)$, for every $M \models T$, is definable by a positive Boolean combination of instances $\varphi(a, y)$ with $a \in M$. In particular, if φ is quantifier-free, then p has a quantifier-free φ -definition.*

2.3. From the order property to tall trees

To prove (3) $\Rightarrow$ (1) of Theorem 1.1.6 we need a supply of many φ -types over a small model. Binary trees (see Definition 2.1.2) provide such a supply.

Proposition 2.3.1. *If $M \models T$ contains an N -ladder for φ , where $N = 2^{n+1} - 1$, then M contains a tree of height n . Hence, if φ has the order property, then φ encodes binary trees of arbitrarily large finite height.*

Proof. For $\sigma \in 2^{\leq n}$ put

$$d(\sigma) = \sum_{k < |\sigma|} (1 - \sigma(k)) 2^{-(k+1)} + 2^{-(|\sigma|+1)} \in (0, 1).$$

The binary expansion of $d(\sigma)$ is $(1 - \sigma(0), \dots, 1 - \sigma(|\sigma| - 1), 1)$, which ends in 1. So d is injective on $2^{\leq n}$, and the image V of d has exactly $|2^{\leq n}| = 2^{n+1} - 1 = N$ elements.

Claim. If $\sigma \frown \langle u \rangle \sqsubseteq \eta$ with $\eta \in 2^n$, then $d(\eta) < d(\sigma) \iff u = 1$.

Indeed, let $k = |\sigma|$. Since $\eta|k = \sigma$ and $\eta(k) = u$,

$$d(\eta) - d(\sigma) = -u 2^{-(k+1)} + R, \quad R = \sum_{k < j < n} (1 - \eta(j)) 2^{-(j+1)} + 2^{-(n+1)}.$$

Here $0 < R \leq (2^{-(k+1)} - 2^{-n}) + 2^{-(n+1)} < 2^{-(k+1)}$. Hence $d(\eta) - d(\sigma)$ is negative if $u = 1$ and positive if $u = 0$, which proves the claim.

Let $a_1, \dots, a_N, b_1, \dots, b_N \in M$ satisfy $\varphi(a_i, b_j) \iff i \leq j$, and let $e: V \rightarrow \{1, \dots, N\}$ be the order isomorphism. Put $b_\sigma := b_{e(d(\sigma))}$ for $\sigma \in 2^{\leq n}$ and $a_\eta := a_{e(d(\eta))}$ for $\eta \in 2^n$. If $\sigma \frown \langle u \rangle \sqsubseteq \eta$, then $d(\eta) \neq d(\sigma)$, and so

$$M \models \varphi(a_\eta, b_\sigma) \iff e(d(\eta)) \leq e(d(\sigma)) \iff d(\eta) < d(\sigma) \iff u = 1.$$

Thus $(b_\sigma), (a_\eta)$ is a tree of height n in M . ☺

2.4. Tall trees produce non-definable types

Lemma 2.4.1 (compactness). *Assume that φ encodes binary trees of arbitrarily large finite height, and let μ be an infinite cardinal. Then there are $M' \models T$ and tuples $(b_s)_{s \in 2^{<\mu}}$ in M' such that for every $\eta \in 2^\mu$ the set*

$$\Sigma_\eta(x) = \{\varphi(x, b_{\eta|_\alpha})^{\eta(\alpha)} : \alpha < \mu\}$$

is finitely satisfiable in M' .

Proof. Add new tuples of constants c_s ($s \in 2^{<\mu}$) and let Γ be T together with the sentences

$$\gamma_{\eta, F} := \exists x \bigwedge_{\alpha \in F} \varphi(x, c_{\eta|_\alpha})^{\eta(\alpha)} \quad (\eta \in 2^\mu, F \subseteq \mu \text{ finite}).$$

By compactness, we need only show that every finite $\Gamma_0 \subseteq \Gamma$ has a model.

Let γ_{η_i, F_i} ($i < m$) be the sentences of this form in Γ_0 (if $m = 0$ there is nothing to do). Let S be the finite set of all $\eta_i|\alpha$ and $\eta_i|(\alpha + 1)$ with $i < m$ and $\alpha \in F_i$. Note that $\alpha + 1 < \mu$ because μ is a limit ordinal. Let $S' = \{s \wedge s' : s, s' \in S\}$, where $s \wedge s'$ is the longest common initial segment. S' is finite and closed under $\wedge$, because in a tree $s \wedge s' \wedge s''$ equals one of the pairwise meets. So S' has a least element r , and every $s \in S' \setminus \{r\}$ has a longest predecessor $s^- \in S'$ with $s^- \sqsubset s$.

Define $f: S' \rightarrow 2^{<N}$, where $N = |S'| + 1$, by recursion on length: $f(r) = \emptyset$, and $f(s) = f(s^-) \frown \langle s(|s^-|) \rangle$. Then $|f(s)| < N$.

Claim. If $s, s' \in S'$ and $s \sqsubset s'$, then $f(s) \frown \langle s'(|s|) \rangle \sqsubseteq f(s')$.

The proof is by induction on $|s'|$. If $s'^- = s$, this is the definition of f . Otherwise $s \sqsubset s'^- \sqsubset s'$ by maximality of s'^- . By induction, $f(s) \frown \langle s'^-(|s|) \rangle \sqsubseteq f(s'^-) \sqsubseteq f(s')$. Finally $s'^-(|s|) = s'(|s|)$.

By hypothesis, take $M_0 \models T$ containing a tree $(b_\sigma)_{\sigma \in 2^{<N}}$, $(a_\nu)_{\nu \in 2^N}$ of height N . Interpret c_s as $b_{f(s)}$ for $s \in S'$, and arbitrarily otherwise. Fix $i < m$, let $\alpha^* = \max F_i$ and $s^* = \eta_i|(\alpha^* + 1) \in S'$. Pick $\nu \in 2^N$ extending $f(s^*)$. For $\alpha \in F_i$ we have $\eta_i|\alpha \sqsubset s^*$ and $s^*(\alpha) = \eta_i(\alpha)$. By the Claim, $f(\eta_i|\alpha) \frown \langle \eta_i(\alpha) \rangle \sqsubseteq \nu$, and so $M_0 \models \varphi(a_\nu, b_{f(\eta_i|\alpha)})^{\eta_i(\alpha)}$. Hence a_ν witnesses γ_{η_i, F_i} , and M_0 with this interpretation is a model of Γ_0 .

By compactness Γ has a model. The L -reduct M' of that model, with $b_s = c_s^{M'}$, is as required. 

Proposition 2.4.2. *Assume that φ encodes binary trees of arbitrarily large finite height. Then for every cardinal $\lambda \geq \kappa$ there is $M \models T$ with $|M| \leq \lambda$ and $|S_\varphi(M)| > \lambda$. In particular, some $p \in S_\varphi(M)$ with $|M| \leq \kappa$ is not definable.*

Proof. Let μ be the least cardinal with $2^\mu > \lambda$. Then μ is infinite and $\mu \leq \lambda$. For $\nu < \mu$ we have $2^\nu \leq \lambda$, so $|2^{<\mu}| = \sum_{\nu < \mu} 2^\nu \leq \mu \cdot \lambda = \lambda$.

Take M' and (b_s) as in Lemma 2.4.1. By the downward Löwenheim–Skolem theorem, there is $M \preceq M'$ with $|M| \leq \lambda$ containing all b_s . Each finite part of Σ_η is satisfiable in M' , hence in M , since $\exists x \bigwedge \dots$ is an $L(M)$ -sentence. So Σ_η is consistent with the elementary diagram of M and extends to some $p_\eta \in S_\varphi(M)$.

If $\eta \neq \eta'$, let α be least with $\eta(\alpha) \neq \eta'(\alpha)$. Then $s = \eta|\alpha = \eta'|\alpha$, and p_η and $p_{\eta'}$ disagree on $\varphi(x, b_s)$. Hence $|S_\varphi(M)| \geq 2^\mu > \lambda$.

For the last statement take $\lambda = \kappa$. A definable type is determined by any φ -definition of the type, and there are only $|L(M)| \leq \kappa$ formulas over M . So at most κ types in $S_\varphi(M)$ are definable, and some $p \in S_\varphi(M)$ is not. 

Proof of (3) $\Rightarrow$ (1) in Theorem 1.1.6. If φ has the order property, then φ encodes binary trees of arbitrarily large finite height by Proposition 2.3.1, and so some φ -type over some model of T is not definable by Proposition 2.4.2. 

This completes the proof of Theorem 1.1.6. We record the tree characterization that came out of this proof.

Theorem 2.4.3. *φ has the order property if and only if φ encodes binary trees of arbitrarily large finite height.*

Proof. The forward direction is Proposition 2.3.1. Conversely, if φ encodes tall trees, then some φ -type is not definable (see Proposition 2.4.2), so φ has the order property by (1) $\Rightarrow$ (3) of Theorem 1.1.6. 

2.5. Historical remarks

- ⌘ **Section 2.2.** Definability of φ -types over models for formulas without the order property is due to Shelah [36, 57]. The sharper form proved here, in which the definition is a positive Boolean combination of instances of φ itself, goes back to Hrushovski and Pillay [19], as Pillay points out in [56, Chapter 1]. Pillay [21, Remark 2.4] derives the positive form directly from Grothendieck's double limit criterion (see Theorem A.1.3 below) by a compactness argument. Positivity matters in applications: for example, positivity gives quantifier-free definitions for quantifier-free formulas.
- ⌘ **Section 2.3.** Binary trees enter stability theory through Shelah's local rank $R(-, \varphi, 2)$. In [36, Lemma 2.7] the rank of a type is computed by the consistency of the family of formulas indexed by the branches of a binary tree of height n . That is, the formulas of infinite 2-rank are exactly those that encode binary trees of arbitrarily large finite height (see Definition 2.1.2). Shelah showed that this happens exactly when φ has the order property [36, 57]. The two directions behave very differently quantitatively. A ladder of length exponential in n yields a tree of height n (see Proposition 2.3.1). In the other direction, a tree yields only a ladder of length logarithmic in the height of the tree; see Section 3.1.
- ⌘ **Section 2.4.** The construction of this section is the local form of Shelah's counting argument [36]. A tree of infinite height gives $2^{\aleph_0}$ φ -types over a countable set of parameters, which is incompatible with definability of all φ -types. This is also the origin of the *stability spectrum*: a theory with the order property has 2^λ types over some set of size λ , for every infinite λ [57].

CHAPTER 3

More stability, trees, and definability

This chapter is a complement to Chapter 2. The proofs given there take a short route, at a price: the way from tall trees back to the order property goes through non-definable types and a count of types over large models, and so says nothing about how tall a tree must be to contain a ladder of a given length. In Section 3.1 we answer that question with a direct combinatorial argument based on Ramsey's theorem. In Section 3.2 we present the classical proof of definability of types, which uses ranks instead of the order property. The conventions are those of Chapter 2.

3.1. A quantitative combinatorial proof (using Ramsey's theorem) that trees give the order property

Theorem 2.4.3 was derived from the definability theorem. Here we give a direct and quantitative proof of the implication “tall trees $\Rightarrow$ order property”. The proof is finite combinatorics: the Boolean case of a Ramsey-type argument due to Hodges. We follow the presentation of Conant and Terry [45].

Definition 3.1.1. Let $s, t \geq 1$.

- (a) A *node embedding* is a map $\theta: 2^{<s} \rightarrow 2^{<t}$ such that $\rho \frown \langle u \rangle \sqsubseteq \tau$ implies $\theta(\rho) \frown \langle u \rangle \sqsubseteq \theta(\tau)$.
- (b) A *full embedding* is a map $\theta: 2^{\leq s} \rightarrow 2^{\leq t}$ sending $2^{<s}$ into $2^{<t}$ and 2^s into 2^t , with the same property for all $\rho, \tau \in 2^{\leq s}$.

Lemma 3.1.2. *Every node embedding $\theta: 2^{<s} \rightarrow 2^{<t}$ extends to a full embedding. If (b_σ) , (a_η) is a tree of height t in M and θ is a full embedding $2^{\leq s} \rightarrow 2^{\leq t}$, then $(b_{\theta(\sigma)} : \sigma \in 2^{<s})$, $(a_{\theta(\eta)} : \eta \in 2^s)$ is a tree of height s in M .*

Proof. For $\eta \in 2^s$, let $\theta(\eta)$ be any element of 2^t extending $\theta(\eta|(s-1)) \frown \langle \eta(s-1) \rangle$. Suppose $\rho \frown \langle u \rangle \sqsubseteq \eta$. If $\rho = \eta|(s-1)$, the requirement holds by construction. Otherwise $\rho \frown \langle u \rangle \sqsubseteq \eta|(s-1)$, so $\theta(\rho) \frown \langle u \rangle \sqsubseteq \theta(\eta|(s-1)) \sqsubseteq \theta(\eta)$. The second statement is immediate, since whether $\varphi(a_\eta, b_\sigma)$ holds depends only on the direction u with $\sigma \frown \langle u \rangle \sqsubseteq \eta$, and full embeddings preserve this direction. ☺

Lemma 3.1.3 (Ramsey lemma for trees). *Let $t_0, t_1 \geq 1$, $t = t_0 + t_1 - 1$, and $2^{<t} = C_0 \cup C_1$. Then for some $u \in \{0, 1\}$ there is a node embedding $\theta: 2^{<t_u} \rightarrow 2^{<t}$ with image contained in C_u .*

Proof. Induction on t . If $t = 1$, then $t_0 = t_1 = 1$, and $\emptyset \mapsto \emptyset$ works for the u with $\emptyset \in C_u$. Let $t > 1$, and choose w with $\emptyset \in C_w$. If $t_w = 1$, the map $\emptyset \mapsto \emptyset$ works, so assume $t_w \geq 2$. For $v \in \{0, 1\}$ and $i \in \{0, 1\}$ let $C_i^v = \{\sigma \in 2^{<t-1} : \langle v \rangle \frown \sigma \in C_i\}$. Then $C_0^v \cup C_1^v$ covers $2^{<t-1}$, and $(t_w - 1) + t_{1-w} - 1 = t - 1$. By induction, for each v either

- (i) there is a node embedding $\chi: 2^{<t_{1-w}} \rightarrow C_{1-w}^v$, in which case $\sigma \mapsto \langle v \rangle \frown \chi(\sigma)$ is a node embedding $2^{<t_{1-w}} \rightarrow 2^{<t}$ with image in C_{1-w} , and we are done; or

(ii) there is a node embedding $\chi_v: 2^{< t_w - 1} \rightarrow C_w^v$.

If (ii) holds for both v , define $\theta: 2^{< t_w} \rightarrow 2^{< t}$ by $\theta(\emptyset) = \emptyset$ and $\theta(\langle v \rangle \frown \sigma) = \langle v \rangle \frown \chi_v(\sigma)$. The image of θ lies in C_w . Suppose $\rho \frown \langle u \rangle \sqsubseteq \tau$. If $\rho = \emptyset$ then $\tau = \langle u \rangle \frown \sigma$ and $\emptyset \frown \langle u \rangle \sqsubseteq \theta(\tau)$. If $\rho = \langle v \rangle \frown \rho'$ then $\tau = \langle v \rangle \frown \tau'$ with $\rho' \frown \langle u \rangle \sqsubseteq \tau'$, so $\chi_v(\rho') \frown \langle u \rangle \sqsubseteq \chi_v(\tau')$. In both cases $\theta(\rho) \frown \langle u \rangle \sqsubseteq \theta(\tau)$, so θ is a node embedding. ☺

Definition 3.1.4. Let $(b_\sigma)_{\sigma \in 2^{< t}}$, $(a_\eta)_{\eta \in 2^t}$ be a tree in M . An *up-ladder of length k* in this tree consists of nodes $\sigma_1, \dots, \sigma_k$ and leaves $\eta_1, \dots, \eta_k$ such that $M \models \varphi(a_{\eta_p}, b_{\sigma_q}) \iff p \leq q$. A *down-ladder of length ℓ* is defined in the same way with “ $p > q$ ” in place of “ $p \leq q$ ”.

Define $B(k, \ell)$ for $k, \ell \geq 1$ by $B(k, 1) = B(1, \ell) = 1$ and $B(k, \ell) = B(k-1, \ell) + B(k, \ell-1) + 1$ for $k, \ell \geq 2$. One checks that $B(k, \ell) = 2^{\binom{k+\ell-2}{k-1}} - 1$.

Proposition 3.1.5. *For all $k, \ell \geq 1$, every tree of height $B(k, \ell)$ for φ in a model of T contains an up-ladder of length k or a down-ladder of length ℓ .*

Proof. Induction on $k + \ell$. In a tree of height 1, the root $\emptyset$ with the leaf $\langle 1 \rangle$ is an up-ladder of length 1, and the root with the leaf $\langle 0 \rangle$ is a down-ladder of length 1. This settles the case $k = 1$ or $\ell = 1$.

Let $k, \ell \geq 2$, $t = B(k, \ell)$, and let $(b_\sigma), (a_\eta)$ be a tree of height t in M . Fix a leaf $\eta_* \in 2^t$ and put

$$P^+ = \{\sigma \in 2^{< t} : M \models \varphi(a_{\eta_*}, b_\sigma)\}, \quad P^- = 2^{< t} \setminus P^+.$$

Let $t^+ = B(k-1, \ell) + 1$ and $t^- = B(k, \ell-1) + 1$, so that $t^+ + t^- - 1 = t$. By Lemma 3.1.3, one of the two cases below occurs.

Case +: there is a node embedding $\theta: 2^{< t^+} \rightarrow 2^{< t}$ with image in P^+ . Extend θ to a full embedding $\hat{\theta}$ (see Lemma 3.1.2) and set $\psi(\tau) = \hat{\theta}(\langle 0 \rangle \frown \tau)$ for $\tau \in 2^{< t^+ - 1}$. Then ψ is a full embedding of height $B(k-1, \ell)$. Put $\sigma_0 = \theta(\emptyset)$. Since $\emptyset \frown \langle 0 \rangle \sqsubseteq \langle 0 \rangle \frown \tau$, we have $\sigma_0 \frown \langle 0 \rangle \sqsubseteq \psi(\tau)$ for every τ . By Lemma 3.1.2, ψ induces a tree of height $B(k-1, \ell)$ in M . By induction, the induced tree contains a down-ladder of length ℓ , which is then a down-ladder in the original tree, or an up-ladder $(\sigma'_p, \eta'_p)_{p=1}^{k-1}$. In the latter case, let

$$(\sigma_1, \eta_1) = (\sigma_0, \eta_*), \quad (\sigma_{p+1}, \eta_{p+1}) = (\psi(\sigma'_p), \psi(\eta'_p)) \quad (1 \leq p < k).$$

We check that $\varphi(a_{\eta_p}, b_{\sigma_q}) \iff p \leq q$.

- (a) For $p = 1$: every σ_q is in the image of θ , which lies in P^+ , so $\varphi(a_{\eta_*}, b_{\sigma_q})$ holds, and indeed $1 \leq q$.
- (b) For $p \geq 2$ and $q = 1$: $\sigma_0 \frown \langle 0 \rangle \sqsubseteq \eta_p$, so $\neg \varphi(a_{\eta_p}, b_{\sigma_0})$, and indeed $p > 1$.
- (c) For $p, q \geq 2$: this is the up-ladder property of the smaller tree.

This is an up-ladder of length k .

Case -: there is a node embedding $\theta: 2^{< t^-} \rightarrow 2^{< t}$ with image in P^- . Proceed symmetrically, with $\psi(\tau) = \hat{\theta}(\langle 1 \rangle \frown \tau)$, a full embedding of height $B(k, \ell-1)$ whose leaves all extend $\sigma_0 \frown \langle 1 \rangle$. By induction the induced tree contains an up-ladder of length k , and we are done, or a down-ladder $(\sigma'_p, \eta'_p)_{p=1}^{\ell-1}$. Prepend (σ_0, η_*) as before.

- (a) For $p = 1$: $\neg \varphi(a_{\eta_*}, b_{\sigma_q})$ for all q , because the nodes lie in P^- , and indeed $1 \not\leq q$.
- (b) For $p \geq 2$ and $q = 1$: $\varphi(a_{\eta_p}, b_{\sigma_0})$ holds, because $\sigma_0 \frown \langle 1 \rangle \sqsubseteq \eta_p$, and indeed $p > 1$.

This is a down-ladder of length ℓ . ☺

Corollary 3.1.6. *If $M \models T$ contains a tree of height $t = B(n+1, n+1)$ for φ , then M contains an n -ladder for φ .*

Proof. By Proposition 3.1.5, M contains an up-ladder or a down-ladder of length $n + 1$. An up-ladder of length $n + 1$ is an $(n + 1)$ -ladder, and so contains an n -ladder. A down-ladder of length $n + 1$ yields an n -ladder by Remark 1.1.5. 

3.2. Optional: The classical proof of definability, via ranks

We give the classical proof that, in the absence of tall trees, every φ -type is definable. Combined with Proposition 2.3.1, this classical proof is an alternative proof of (1) $\Rightarrow$ (3) in Theorem 1.1.6. The definitions obtained in this section, however, are not positive Boolean combinations of instances of φ : they contain quantifiers and negations that do not occur in φ (see Lemma 3.2.2(iv)). Assume that there is n such that no model of T contains a tree of height n , and fix $M \models T$. By a φ -conjunction over M we mean a finite conjunction $\theta(x)$ of formulas $\varphi(x, b)^u$ with $b \in M$.

Definition 3.2.1. For a φ -conjunction θ over M and $m < \omega$, write $R(\theta) \geq m$ if there are $b_\sigma \in M$ ($\sigma \in 2^{<m}$) such that for every $\eta \in 2^m$,

$$M \models \exists x \left(\theta(x) \wedge \bigwedge_{k < m} \varphi(x, b_{\eta|k})^{\eta(k)} \right).$$

Thus $R(\theta) \geq 0$ if and only if $M \models \exists x \theta(x)$. Let $R(\theta)$ be the largest m with $R(\theta) \geq m$.

Lemma 3.2.2.

- (i) $R(\theta) < n$ for every θ ; in particular $R(\theta)$ is well defined whenever θ is satisfiable in M .
- (ii) If $\theta' = \theta \wedge \chi$, then $R(\theta') \leq R(\theta)$.
- (iii) If $b \in M$ and $R(\theta \wedge \varphi(x, b)^u) \geq m$ for both $u \in \{0, 1\}$, then $R(\theta) \geq m + 1$.
- (iv) For each θ and m there is an $L(M)$ -formula $\psi_{\theta, m}(y)$ such that for all $b \in M$, $R(\theta \wedge \varphi(x, b)) \geq m \iff M \models \psi_{\theta, m}(b)$.

Proof. (i) If $R(\theta) \geq n$, witnessed by (b_σ) , then choosing a witness $a_\eta \in M$ for each $\eta \in 2^n$ gives a tree of height n in M , which is impossible.

(ii) A witness for $R(\theta') \geq m$ is a witness for $R(\theta) \geq m$.

(iii) Let $(b_\sigma^u)_{\sigma \in 2^{<m}}$ witness $R(\theta \wedge \varphi(x, b)^u) \geq m$. Put $b_\emptyset = b$ and $b_{\langle u \rangle \frown \sigma} = b_\sigma^u$. A branch of length $m + 1$ has the form $\langle u \rangle \frown \eta$ with $\eta \in 2^m$, and the corresponding formula is $\theta \wedge \varphi(x, b)^u \wedge \bigwedge_{k < m} \varphi(x, b_{\eta|k}^u)^{\eta(k)}$, which is satisfiable in M .

(iv) Take

$$\psi_{\theta, m}(y) := \exists (z_\sigma)_{\sigma \in 2^{<m}} \bigwedge_{\eta \in 2^m} \exists x \left(\theta(x) \wedge \varphi(x, y) \wedge \bigwedge_{k < m} \varphi(x, z_{\eta|k})^{\eta(k)} \right),$$

whose parameters are those of θ . 

Proposition 3.2.3. If for some n no model of T contains a tree of height n , then every $p \in S_\varphi(M)$ is definable.

Proof. Let Φ_p be the set of finite conjunctions of members of p . Each $\theta \in \Phi_p$ is satisfiable in M : $\exists x \theta$ is an $L(M)$ -sentence consistent with the complete theory of $(M, b)_{b \in M}$, hence true in M . By Lemma 3.2.2(i), choose $\theta \in \Phi_p$ with $m = R(\theta)$ minimal. We show that $\psi_{\theta, m}$ is a φ -definition of p , i.e. that for $b \in M$,

$$\varphi(x, b) \in p \iff R(\theta \wedge \varphi(x, b)) \geq m.$$

If $\varphi(x, b) \in p$, then $\theta \wedge \varphi(x, b) \in \Phi_p$, so $R(\theta \wedge \varphi(x, b)) \geq m$ by minimality. Conversely, suppose $\varphi(x, b) \notin p$. Then $\neg \varphi(x, b) \in p$, so $R(\theta \wedge \neg \varphi(x, b)) \geq m$ by minimality. If also

$R(\theta \wedge \varphi(x, b)) \geq m$, then Lemma 3.2.2(iii) would give $R(\theta) \geq m + 1$, a contradiction. The equivalence together with Lemma 3.2.2(iv) shows that p is definable. ☺



Remark 3.2.4.

- (a) The arguments are effective. An up- or down-ladder of length $n + 1$ appears in any tree of height $B(n + 1, n + 1) = \binom{2n+2}{n+1} - 1$. Conversely, a ladder of length $2^{n+1} - 1$ yields a tree of height n .
- (b) Theorem 2.2.1 uses only that p is finitely satisfiable in M (see Lemma 1.1.2). So the theorem applies to any φ -type over a set A that is finitely satisfiable in A . The parameters of that definition come from the finite set F , whereas the rank argument of Proposition 3.2.3 yields a definition whose parameters are those of the formula θ of minimal rank.
- (c) Without finite satisfiability, definitions over the parameter set cannot be expected in general, and the restriction to types over models is essential. Definition 3.2.5 and Example 3.2.6 below make this precise.

Definition 3.2.5. Let $N \models T$, $A \subseteq N$, and $p \in S_\varphi(N)$.

- (a) A *complete φ -type over A* is defined as in Definition 1.1.1, with b ranging over A instead of M ; such a type is *finitely satisfiable in A* if every finite subset of the type is realized by an element of A .
- (b) p is *definable over A* if p has a φ -definition that is an $L(A)$ -formula.
- (c) p is *definable almost over A* if p has a φ -definition of the form $\psi(y, c)$, with $\psi(y, z)$ an L -formula and c a tuple of N , such that only finitely many distinct subsets of N are of the form $\psi(N, c')$ with $c' \in N$ realizing the same complete type over A as c . In other words, the definition of p has only finitely many conjugates over A .

Definable over A implies definable almost over A : take c to enumerate the parameters from A , so that $c' = c$ is the only possibility. The converse fails.

Example 3.2.6. Let T be the theory of an equivalence relation E with exactly two classes, both infinite, let $\varphi(x, y)$ be $E(x, y)$, and let $A = \emptyset$. Let $N \models T$ with classes C_1 and C_2 , and let $p \in S_\varphi(N)$ be the φ -type saying that x lies in C_1 , i.e. $E(x, b) \in p$ if and only if $b \in C_1$. For any $c \in C_1$, the formula $E(y, c)$ is a φ -definition of p . All elements of N have the same complete type over $\emptyset$, so the sets $E(N, c')$ are exactly C_1 and C_2 . Hence p is definable almost over $\emptyset$. The type p is not definable over $\emptyset$: since N has an automorphism swapping C_1 and C_2 , every L -formula $\psi(y)$ without parameters defines $\emptyset$ or N , never C_1 . Note that p is not finitely satisfiable in $\emptyset$.

3.3. Historical remarks



Section 3.1. The Ramsey-type argument that extracts ladders from trees is due to Hodges [53]. Hodges's argument gives a direct combinatorial proof that tall trees force the order property, without going through definability of types. Conant and Terry [45] recently revisited the argument with explicit bounds, and extended the argument from $\{0, 1\}$ -valued formulas to real-valued functions. Their version is the one followed here. Such quantitative bounds are what is needed for applications of stability to finite combinatorics, such as the regularity lemma for stable graphs.



Section 3.2. The rank $R(\theta)$ is Shelah's local rank $R(\theta, \varphi, 2)$ [36, 57]. The proof that a type of minimal rank is definable is the classical route to definability of types, the one found in most textbook treatments, for instance Pillay [56] and Tent–Ziegler [58], and predates

the combinatorial and analytic proofs given elsewhere in these notes. As noted above, the resulting definitions involve quantifiers, whereas those of Section 2.2 do not.

CHAPTER 4

Stability and notions of independence

Stability can also be understood through independence. In this chapter we show that the stable theories are exactly those carrying a notion of independence, written $A \downarrow_M B$ and read “ A is independent from B over M ”, that obeys a short list of natural axioms. The most important of these are finite character, symmetry, and the existence and uniqueness of independent extensions. The idea is that, given a type p over a model M and a set B , an independent extension of p to $M \cup B$ adds no information beyond p . Stability is precisely the condition that such an extension always exists and is unique.

Both directions rest on Chapter 2. To obtain independence from stability we use definability of types: the independent extension is the *heir*, the extension of p given by the same definition scheme as p . The converse is a counting argument of the kind used there.

4.1. Conventions

From now on T is a *complete* theory in the language L , and $\kappa = |T| = |L| + \aleph_0$. We work inside a monster model $\mathbb{M}$ of T : a $\bar{\kappa}$ -saturated and strongly $\bar{\kappa}$ -homogeneous model for some very large cardinal $\bar{\kappa}$ (larger than 2^{2^κ} , say). A set is *small* if of cardinality $< \bar{\kappa}$. Letters A, B, C denote small subsets of $\mathbb{M}$; tuples may be infinite, but are always small. We freely identify a set with an enumerating tuple. A *model* is a small elementary substructure $M \preceq \mathbb{M}$. Every model of T of size $< \bar{\kappa}$ embeds elementarily in $\mathbb{M}$, and every type over a small set is realized in $\mathbb{M}$.

We write $\text{tp}(a/A)$ for the complete type of a over A , $S_x(A)$ for the set of complete types over A in the variables x , and $a \equiv_A a'$ for $\text{tp}(a/A) = \text{tp}(a'/A)$. We write MB for $M \cup B$. By strong homogeneity, $a \equiv_A a'$ if and only if $\sigma(a) = a'$ for some $\sigma \in \text{Aut}(\mathbb{M}/A)$.

Definition 4.1.1.

- (a) T is *stable* if no L -formula $\varphi(x, y)$ has the order property (see Definition 1.1.4).
- (b) For an infinite cardinal λ , T is λ -*stable* if $|S_x(A)| \leq \lambda$ whenever $|A| \leq \lambda$ and x is a finite tuple of variables.

Lemma 4.1.2. *If T is λ -stable for some $\lambda \geq \kappa$, then T is stable.*

Proof. Suppose that $\varphi(x, y)$ has the order property. By Proposition 2.3.1, φ encodes binary trees of arbitrarily large finite height. By Proposition 2.4.2 there is $N \models T$ with $|N| \leq \lambda$ and $|S_\varphi(N)| > \lambda$. Since T is complete, we may assume $N \preceq \mathbb{M}$. Every φ -type over N is consistent with the elementary diagram of N , so each φ -type over N extends to a complete type in $S_x(N)$, and distinct φ -types have distinct extensions. Hence $|S_x(N)| > \lambda \geq |N|$, and T is not λ -stable. ☺

4.2. Definitions and heirs

Lemma 4.2.1. *Assume T is stable, and let $p \in S_x(M)$, where x may be infinite. For every $L(M)$ -formula $\varphi(x', y)$, with x' a finite subtuple of x and y finite, there is an $L(M)$ -formula*

$d_p\varphi(y)$ such that for all $b \in M$,

$$\varphi(x', b) \in p \iff M \models d_p\varphi(b).$$

Any two formulas with this property are equivalent in $\mathbb{M}$. We call $(d_p\varphi)_\varphi$ the definition scheme of p .

Proof. Write $\varphi(x', y) = \varphi_0(x', y, m)$ with $\varphi_0 \in L$ and $m \in M$. The set of formulas $\varphi_0(x', c)^u$ ($c \in M$) that belong to p is a complete φ_0 -type over M in the sense of Definition 1.1.1. Since φ_0 does not have the order property, Theorem 1.1.6 shows that this φ_0 -type has a definition $\psi(z)$ over M , which is a positive Boolean combination of instances of φ_0 . Here z corresponds to the parameter variables y followed by those replaced by m . Then $d_p\varphi(y) := \psi(y, m)$ works. If $\psi_1(y)$ and $\psi_2(y)$ both work, then $M \models \forall y (\psi_1 \leftrightarrow \psi_2)$. This is an $L(M)$ -sentence, so this sentence also holds in $\mathbb{M}$. $\odot$

Definition 4.2.2. Assume T is stable. Let $p \in S_x(M)$ and let B be small. Define

$$p|B := \{\varphi(x', b) : \varphi(x', y) \in L(M), b \in B, \mathbb{M} \models d_p\varphi(b)\}.$$

Every $L(MB)$ -formula has the form $\varphi(x', b)$ with $\varphi \in L(M)$ and $b \in B$, so $p|B$ is a set of $L(MB)$ -formulas.

Lemma 4.2.3. Assume T is stable and let $p \in S_x(M)$.

- (a) $p|B \in S_x(MB)$, and $p|B$ extends p .
- (b) If $B \subseteq D$, then $p|B$ is the restriction of $p|D$ to MB .
- (c) If N is a model with $M \subseteq N$, then $p|N$ has the same definition scheme as p , and $(p|N)|B = p|NB$ for every small B .

Proof. (a) *Completeness.* Let $\varphi \in L(M)$. Since p is complete, for every $c \in M$ exactly one of $\varphi(x', c)$, $\neg\varphi(x', c)$ lies in p . Hence $M \models \forall y (d_p(\neg\varphi)(y) \leftrightarrow \neg d_p\varphi(y))$, and the same holds in $\mathbb{M}$. So for $b \in B$ exactly one of $\varphi(x', b)$, $\neg\varphi(x', b)$ belongs to $p|B$.

Consistency. Let $\varphi_i(x_i, b_i) \in p|B$ for $i < n$, and put $\chi(\bar{x}; y_0, \dots, y_{n-1}) = \bigwedge_{i < n} \varphi_i(x_i, y_i)$, where $\bar{x}$ lists the variables in the x_i . For $\bar{c} \in M$, the formula $\chi(\bar{x}, \bar{c})$ is in p if and only if each $\varphi_i(x_i, c_i)$ is, and in that case $\chi(\bar{x}, \bar{c})$ is satisfiable in M . Therefore

$$M \models \forall \bar{y} \left(\bigwedge_{i < n} d_p\varphi_i(y_i) \rightarrow d_p\chi(\bar{y}) \right) \quad \text{and} \quad M \models \forall \bar{y} (d_p\chi(\bar{y}) \rightarrow \exists \bar{x} \chi(\bar{x}, \bar{y})).$$

Both sentences transfer to $\mathbb{M}$. Applied to $\bar{b} = (b_0, \dots, b_{n-1})$, they give $\mathbb{M} \models \exists \bar{x} \bigwedge_i \varphi_i(x_i, b_i)$. By compactness, $p|B$ is consistent.

Extension. For $c \in M$, $\varphi(x', c) \in p \iff M \models d_p\varphi(c)$, which holds if and only if $\varphi(x', c) \in p|B$.

(b) This is immediate from the definition.

(c) By (a), $p|N$ extends p . For $n \in N$ we have $\varphi(x', n) \in p|N$ if and only if $\mathbb{M} \models d_p\varphi(n)$, so $d_p\varphi$ is a definition of $p|N$ for $\varphi \in L(M)$. For an $L(N)$ -formula $\varphi(x', y) = \varphi_0(x', y, n)$ with $\varphi_0 \in L(M)$, the formula $d_p\varphi_0(y, n)$ defines $p|N$. By the uniqueness clause of Lemma 4.2.1, the definition scheme of $p|N$ is thus given, up to equivalence, by that of p . Now for $b \in B$ and $n \in N$,

$$\varphi_0(x', b, n) \in (p|N)|B \iff \mathbb{M} \models d_p\varphi_0(b, n) \iff \varphi_0(x', b, n) \in p|NB. \quad \odot$$

The following lemma, which says that definitions of types over a model commute, is the heart of the matter, and the only place in this chapter where the absence of the order property is used directly.

Lemma 4.2.4 (Harrington's lemma). *Assume T is stable. Let $p \in S_x(M)$ and $q \in S_y(M)$ with x, y finite, and let $\varphi(x, y) \in L(M)$. Write $d_p\varphi(y)$ for the definition of p for $\varphi(x, y)$. Write $d_q\varphi(x)$ for the definition of q for φ , regarded as a formula in the variables y with parameter variables x . Then*

$$d_p\varphi(y) \in q \iff d_q\varphi(x) \in p.$$

Proof. Choose recursively $a_i, b_i \in \mathbb{M}$ ($i < \omega$) such that a_i realizes $p|_{\{b_j : j < i\}}$ and b_i realizes $q|_{\{a_j : j \leq i\}}$. This is possible by Lemma 4.2.3(a) and saturation. In particular $a_i \models p$ and $b_i \models q$.

If $j < i$, then since $a_i \models p|_{\{b_j\}}$ and $b_j \models q$,

$$\mathbb{M} \models \varphi(a_i, b_j) \iff \mathbb{M} \models d_p\varphi(b_j) \iff d_p\varphi(y) \in q.$$

If $i \leq j$, then since $b_j \models q|_{\{a_i\}}$ and $a_i \models p$,

$$\mathbb{M} \models \varphi(a_i, b_j) \iff \mathbb{M} \models d_q\varphi(a_i) \iff d_q\varphi(x) \in p.$$

Suppose the two right-hand sides differ, and write $\varphi(x, y) = \varphi_0(x, y, m)$ with $\varphi_0 \in L$ and $m \in M$. Put $b'_j = b_j m$. Then either $\mathbb{M} \models \varphi_0(a_i, b'_j) \iff i \leq j$ for all i, j , or $\mathbb{M} \models \varphi_0(a_i, b'_j) \iff i > j$ for all i, j . In the first case $\varphi_0(x; y, z)$ has the order property directly, and in the second by Remark 1.1.5. Either way this contradicts stability. $\odot$

4.3. Independence relations

Definition 4.3.1. An *independence relation (over models)* is a relation $A \downarrow_M B$ between small sets A, B and models M satisfying the following axioms, for all small A, A', B, B' and all models M, N .

- (INV) *Invariance.* If $A \downarrow_M B$ and $\sigma \in \text{Aut}(\mathbb{M})$, then $\sigma(A) \downarrow_{\sigma(M)} \sigma(B)$.
- (MON) *Monotonicity.* If $A \downarrow_M B$, $A' \subseteq A$ and $B' \subseteq B$, then $A' \downarrow_M B'$.
- (BMON) *Base monotonicity.* If $A \downarrow_M B$ and N is a model with $M \subseteq N \subseteq B$, then $A \downarrow_N B$.
- (TR) *Transitivity.* If $M \subseteq N$, $A \downarrow_M N$ and $A \downarrow_N B$, then $A \downarrow_M NB$.
- (FC) *Finite character.* $A \downarrow_M B$ if and only if $a \downarrow_M b$ for all finite tuples a from A and b from B .
- (SYM) *Symmetry.* $A \downarrow_M B$ if and only if $B \downarrow_M A$.
- (EX) *Existence (extension).* For all A, M, B there is $A' \equiv_M A$ with $A' \downarrow_M B$.
- (UN) *Uniqueness (stationarity).* If $A \equiv_M A'$, $A \downarrow_M B$ and $A' \downarrow_M B$, then $A \equiv_{MB} A'$.
- (LOC) *Local character.* For every finite tuple a and every model N there is a model $M \subseteq N$ with $|M| \leq \kappa$ and $a \downarrow_M N$.

In terms of types, (EX) and (UN) say that a type $p \in S_x(M)$ has exactly one *independent extension* to MB , namely the type over MB of any $A \models p$ with $A \downarrow_M B$. All the other axioms are statements about which extensions of p count as independent.

Definition 4.3.2. Assume T is stable. For small A, B and a model M we set

$$A \downarrow_M B \quad : \iff \quad \text{tp}(A/MB) = \text{tp}(A/M)|_B.$$

In words, $\text{tp}(A/MB)$ is the *heir* of $\text{tp}(A/M)$, the extension defined by the same definition scheme.

For finite tuples a, b with $p = \text{tp}(a/M)$, every $L(Mb)$ -formula in x has the form $\varphi(x, b)$ with $\varphi(x, y) \in L(M)$. So

$$(4.1) \quad a \downarrow_M b \iff \text{for all } \varphi(x, y) \in L(M) : \quad \mathbb{M} \models \varphi(a, b) \leftrightarrow d_p\varphi(b).$$

Theorem 4.3.3. *For a complete theory T the following are equivalent.*

- (1) T is stable.
- (2) There is an independence relation satisfying all the axioms of Definition 4.3.1.
- (3) There is a relation $A \downarrow_M B$ satisfying (UN) and (LOC).

Moreover, if T is stable, the relation of Definition 4.3.2 satisfies all the axioms. This relation also satisfies

(AR) Anti-reflexivity. If $a \downarrow_M a$ for a finite tuple a , then $a \subseteq M$.

4.3.1. Proof of (1) $\Rightarrow$ (2). Assume T is stable, and let $\downarrow$ be the relation of Definition 4.3.2. Throughout, $p = \text{tp}(A/M)$.

(INV). Let $\sigma \in \text{Aut}(\mathbb{M})$. Applying σ to the parameters of the definition scheme of p gives a definition scheme for $\sigma(p) = \text{tp}(\sigma A/\sigma M)$. Hence $\sigma(p)|\sigma(B) = \sigma(p|B)$. If $\text{tp}(A/MB) = p|B$, then $\text{tp}(\sigma A/\sigma M \sigma B) = \sigma(p|B) = \sigma(p)|\sigma B$.

(MON). The definition scheme of $\text{tp}(A'/M)$ is the part of that of p that involves only variables for A' . So $\text{tp}(A'/M)|B'$ is the restriction of $p|B$ to these variables and to MB' (see Lemma 4.2.3(b)), which is $\text{tp}(A'/MB')$.

(FC). The condition $\text{tp}(A/MB) = p|B$ says that for every $L(M)$ -formula $\varphi(x', y)$ and every $b \in B$ we have $\mathbb{M} \models \varphi(a', b) \leftrightarrow d_p \varphi(b)$, where a' is the finite subtuple of A corresponding to x' . Each instance involves only finite $a' \subseteq A$ and $b \subseteq B$, and is exactly an instance of $a' \downarrow_M b$.

(BMON). Let $M \subseteq N \subseteq B$ and $\text{tp}(A/MB) = p|B$. Then $\text{tp}(A/N) = p|N$ by Lemma 4.2.3(b). By Lemma 4.2.3(c), $\text{tp}(A/N)|B = (p|N)|B = p|NB = p|B = \text{tp}(A/NB)$, so $A \downarrow_N B$.

(TR). Let $M \subseteq N$, $\text{tp}(A/N) = p|N$ and $\text{tp}(A/NB) = \text{tp}(A/N)|B$. By Lemma 4.2.3(c), $\text{tp}(A/NB) = (p|N)|B = p|NB$, i.e. $A \downarrow_M NB$.

(SYM). By (FC) we need only treat finite tuples a, b . Let $p = \text{tp}(a/M)$, $q = \text{tp}(b/M)$, and assume $a \downarrow_M b$. For $\varphi(x, y) \in L(M)$, using (4.1) and Lemma 4.2.4,

$$\mathbb{M} \models \varphi(a, b) \iff \mathbb{M} \models d_p \varphi(b) \iff d_p \varphi(y) \in q \iff d_q \varphi(x) \in p \iff \mathbb{M} \models d_q \varphi(a).$$

By (4.1) with the roles of a and b exchanged, this says $b \downarrow_M a$.

(EX). By Lemma 4.2.3(a) and saturation, $p|B$ has a realization A' . Then $A' \equiv_M A$ and $A' \downarrow_M B$.

(UN). If $A \equiv_M A'$, then both have type p over M . If both are independent from B over M , then $\text{tp}(A/MB) = p|B = \text{tp}(A'/MB)$.

(LOC). Let a be finite, N a model and $p = \text{tp}(a/N)$. For each L -formula $\varphi(x, y)$ fix a definition $d_p \varphi(y) \in L(N)$, and let $D \subseteq N$ be the set of all parameters used, so $|D| \leq \kappa$. By downward Löwenheim–Skolem there is a model $M \preceq N$ with $D \subseteq M$ and $|M| \leq \kappa$. Each $d_p \varphi$ is an $L(M)$ -formula, and on M the formula $d_p \varphi$ defines $q = \text{tp}(a/M)$ for φ . So these formulas form a definition scheme for q (for $L(M)$ -formulas $\varphi_0(x, y, m)$ use $d_p \varphi_0(y, m)$). Hence $q|N$ is defined by the same formulas as p . Since $q|N$ and p are both complete types over N , $q|N = p = \text{tp}(a/N)$, i.e. $a \downarrow_M N$.

(AR). Suppose $a \downarrow_M a$ with $p = \text{tp}(a/M)$, and suppose some coordinate a_0 of a is not in M . Let $\varphi(x, y)$ be the formula $x_0 = y_0$, where x_0 and y_0 are the corresponding coordinates of x and y . For $c \in M$ we have $\varphi(x, c) \notin p$, because $a_0 \neq c_0$. So $M \models \neg \exists y d_p \varphi(y)$, and the same holds in $\mathbb{M}$. By (4.1), $\mathbb{M} \models \varphi(a, a) \leftrightarrow d_p \varphi(a)$. But $\varphi(a, a)$ holds and $d_p \varphi(a)$ fails, a contradiction.

4.3.2. Proof of (2) $\Rightarrow$ (3). This is trivial.

4.3.3. Proof of (3) $\Rightarrow$ (1). Let $\perp$ satisfy (UN) and (LOC), and put $\lambda = 2^\kappa$, so that $\lambda^\kappa = \lambda$. We show that T is λ -stable. Then T is stable by Lemma 4.1.2.

Let $|A| \leq \lambda$ and let x be finite. By downward Löwenheim–Skolem there is a model $N \supseteq A$ with $|N| \leq \lambda$, and restriction maps $S_x(N)$ onto $S_x(A)$. So we need only show $|S_x(N)| \leq \lambda$.

For each $p \in S_x(N)$ choose a realization a_p . By (LOC) choose a model $M_p \subseteq N$ with $|M_p| \leq \kappa$ and $a_p \perp_{M_p} N$. Consider the map

$$p \mapsto (M_p, \text{tp}(a_p/M_p)).$$

This map is injective. Indeed, suppose $M_p = M_{p'} =: M$ and $a_p \equiv_M a_{p'}$. Since $a_p \perp_M N$ and $a_{p'} \perp_M N$, (UN) gives $a_p \equiv_N a_{p'}$, i.e. $p = p'$.

The number of subsets of N of size $\leq \kappa$ is at most $\lambda^\kappa = \lambda$. For each such M , $|S_x(M)| \leq 2^\kappa = \lambda$. Hence $|S_x(N)| \leq \lambda \cdot \lambda = \lambda$. 

4.4. Remarks

 **Remark 4.4.1.** The proof of (3) $\Rightarrow$ (1) uses only uniqueness and local character. Local character may be weakened to $|M| < \kappa_0$ for any fixed cardinal κ_0 , by taking $\lambda = 2^{\kappa_0 + \kappa}$. So stability is equivalent to the existence of a relation for which types over models have unique independent extensions and every type is independent over a small submodel. All the other axioms come for free once T is stable.

 **Remark 4.4.2.** In a stable theory the relation of Definition 4.3.2 is *nonforking independence* restricted to models as bases. For $M \subseteq B$, the following are equivalent: $\text{tp}(a/B)$ does not fork over M ; $\text{tp}(a/B)$ is the heir of $\text{tp}(a/M)$; $\text{tp}(a/B)$ is definable over M ; see Pillay [56] or Tent–Ziegler [58]. Nonforking makes sense over arbitrary base sets C . Over such sets, nonforking satisfies all the axioms above except uniqueness, which holds only over algebraically closed sets in T^{eq} (Harnik–Harrington [54]). Kim and Pillay [55] proved that in a stable (indeed in a simple) theory, nonforking is the *only* relation on arbitrary small sets satisfying invariance, monotonicity, transitivity, finite character, symmetry, extension, local character and the independence theorem. Developing forking over arbitrary sets requires considerably more machinery, which is why we have taken models as base sets.

4.5. Historical remarks

 **Abstract independence.** The axiomatic study of independence is older than model theory. Steinitz, in his foundational paper on the algebraic theory of fields [59], introduced transcendence bases and showed that any two transcendence bases have the same cardinality. The argument is an exchange argument of the kind now called the Steinitz exchange lemma, the same one that gives the dimension of a vector space. Van der Waerden, in *Moderne Algebra* [60], observed that linear dependence and algebraic dependence obey the same few rules, and derived the existence of bases and the invariance of their cardinality from an abstract notion of dependence whose central axiom is exchange. Whitney [61] went a step further. He introduced *matroids*, structures that isolate the abstract properties of linear dependence, and gave several equivalent axiomatizations, in terms of independent sets, bases, circuits and rank. Nakasawa developed the same theory independently in the mid-1930s, and Birkhoff [62] and Mac Lane [63] related matroid theory to lattices and to projective geometry.

After a period in which matroids were studied mainly by Rado and Tutte, Rota made them central to combinatorics. In his monograph with Crapo [64] they become *combinatorial geometries*, and a closure operator with the exchange property is called a *pregeometry*, the term still used in model theory. Rota’s view was that dependence is a geometric notion with

invariants of its own, such as the Möbius function and the characteristic polynomial of the associated lattice.

Model theory met these ideas in the study of uncountably categorical theories. Baldwin and Lachlan [65] showed that in a strongly minimal set algebraic closure has the exchange property, so algebraic closure is a pregeometry and every model has a well-defined dimension. They used this to give a new proof of Morley’s categoricity theorem [35]. Shelah’s forking [57] carries independence far beyond pregeometries: nonforking independence is not given by a closure operator, yet in a stable theory nonforking independence has the symmetry, existence and uniqueness properties on which arguments about dimension depend. The later work of Zilber and Hrushovski on the geometry of forking turned the study of forking into a subject of its own, geometric stability theory [56]. The point of view of this chapter, which characterizes independence axiomatically, as van der Waerden and Whitney did for linear dependence, is that of Harnik and Harrington and of Kim and Pillay, discussed below.

- ✂ **Section 4.1.** Working inside a single large saturated and strongly homogeneous model goes back to Shelah [57], who wrote $\mathfrak{C}$ for this model. The name *monster model* became standard later. The notion of ω -stability comes from Morley’s categoricity theorem [35]. The study of λ -stability for arbitrary λ , including the fact that stability is equivalent to λ -stability for all λ with $\lambda^{|T|} = \lambda$, is due to Shelah [36, 57].
- ✂ **Section 4.2.** Heirs and coheirs (*héritiers* and *cohéritiers*) were introduced by Lascar and Poizat [33]. They gave an alternative presentation of Shelah’s forking in stable theories, based only on definability of types and finite satisfiability. Lemma 4.2.4 says that the definitions of two types over a model commute. Lemma 4.2.4 is due to Harrington, appears in Harnik–Harrington [54], and is the key step in proving symmetry of independence in stable theories.
- ✂ **Section 4.3.** Nonforking independence was introduced by Shelah [57]. Lascar and Poizat [33] reformulated nonforking independence in terms of heirs and definability. The characterization of stability through abstract independence notions is due to Harnik and Harrington [54]. Kim and Pillay [55] later extended this characterization from stable to simple theories. They proved that the simple theories are exactly those carrying a relation satisfying the axioms listed above, with the independence theorem in place of uniqueness, and that such a relation is then nonforking. Theorem 4.3.3 is the stable case over models, with uniqueness (stationarity over models) as the key axiom. The anchor symbol $\perp$ for independence was introduced by Makkai in a lecture in Jerusalem, during the Model Theory Year at the Institute for Advanced Studies (1980–81), and was adopted at once. The symbol appears in print in Makkai’s survey [48].

CHAPTER 5

From stability to NIP

In a stable theory no formula can encode arbitrarily long ladders. Many of the best understood structures in mathematics fail this test for a simple reason: they carry a definable linear order. Dense linear orders, the ordered field of real numbers and, more generally, o-minimal structures are all unstable, and yet their definable sets are remarkably well behaved. To account for this kind of tameness we introduce the *independence property*, a configuration much wilder than a ladder, and study the formulas and theories that avoid the independence property, called *NIP*. Every stable formula is NIP, and so are the orders just mentioned, so NIP generalizes both stability and the tameness of order. In Section 5.1 we keep the conventions of Section 1.1, and in Section 5.2 those of Section 1.2.

5.1. The independence property

A ladder (see Definition 1.1.4) is a special pattern: the element a_i satisfies $\varphi(x, b_j)$ exactly for $j \geq i$. The independence property asks instead that *every* pattern be realized.

Definition 5.1.1. Let $M \models T$ and $n \geq 1$.

- (a) Tuples $b_1, \dots, b_n$ of M are *shattered by φ in M* if for every $S \subseteq \{1, \dots, n\}$ there is a tuple a_S of M such that, for all $1 \leq i \leq n$,

$$M \models \varphi(a_S, b_i) \iff i \in S.$$

- (b) φ has the *independence property* if for every n some model of T contains n tuples shattered by φ . Otherwise φ is *NIP* (“not the independence property”); NIP formulas are also called *dependent*.
- (c) The *VC dimension* of φ is the largest n such that some model of T contains n tuples shattered by φ , and ∞ if there is no largest such n . Thus φ is NIP if and only if the VC dimension of φ is finite.

A theory is *NIP* if every formula is NIP.

In the terminology of learning theory, the family of sets $\{\varphi(a, M) : a \in M\}$, where $\varphi(a, M) = \{b \in M : M \models \varphi(a, b)\}$, shatters $\{b_1, \dots, b_n\}$: every subset is cut out by one of the sets in the family. As for ladders, an $(n+1)$ -shattered family contains an n -shattered one, so by compactness φ has the independence property if and only if some model of T contains an infinite sequence $(b_i)_{i < \omega}$ such that for every $S \subseteq \omega$ there is a_S with $\varphi(a_S, b_i)$ if and only if $i \in S$.

Proposition 5.1.2. *If φ does not have the order property, then φ is NIP.*

Proof. Suppose $b_1, \dots, b_n$ are shattered by φ in M , witnessed by the tuples a_S . For $1 \leq i \leq n$ put $S_i = \{j : i \leq j \leq n\}$ and $a_i = a_{S_i}$. Then $M \models \varphi(a_i, b_j)$ if and only if $j \in S_i$, that is, if and only if $i \leq j$. So $a_1, \dots, a_n$ and $b_1, \dots, b_n$ form an n -ladder. If φ has the independence property, then φ also has the order property. ☺

The converse fails, and the first example of the chapter shows why NIP is the right notion for ordered structures.

Example 5.1.3. Let T be the theory of $(\mathbb{Q}, <)$ and let $\varphi(x, y)$ be $x < y$. Then $a_i = i$ and $b_j = j + \frac{1}{2}$ satisfy $\varphi(a_i, b_j)$ if and only if $i \leq j$, so φ has the order property and is unstable. But φ is NIP, with VC dimension 1. Indeed, let b_1, b_2 be elements of a model, say $b_1 \leq b_2$. Shattering requires some a with $a < b_1$ and $a \not< b_2$, that is, $b_2 \leq a < b_1$, which is impossible. So no two elements are shattered. A single element b is shattered, by any $a < b$ and any $a \geq b$.

Example 5.1.4. Let T be the theory of the random graph, whose models are the graphs in which, for all disjoint finite sets U, V of vertices, some vertex outside $U \cup V$ is adjacent to every vertex of U and to no vertex of V . Let $\varphi(x, y)$ be the edge relation $E(x, y)$. Any n distinct vertices $b_1, \dots, b_n$ are shattered: for $S \subseteq \{1, \dots, n\}$ apply the extension property to $U = \{b_i : i \in S\}$ and $V = \{b_i : i \notin S\}$. So E has the independence property.

The independence property of $\varphi(x, y)$ refers to shattering tuples on the y -side. Which side is used does not matter (see Proposition 5.1.5 below). Write $\varphi^{\text{opp}}(y, x)$ for $\varphi(x, y)$ with the roles of the variables exchanged.

Proposition 5.1.5. φ is NIP if and only if φ^{opp} is NIP. More precisely, if some model contains 2^n tuples shattered by φ , then that model contains n tuples shattered by φ^{opp} .

Proof. Index the 2^n tuples as b_T , $T \subseteq \{1, \dots, n\}$, and let a_S , $S \subseteq \mathcal{P}(\{1, \dots, n\})$, witness that they are shattered by φ . For $1 \leq i \leq n$ put $S_i = \{T : i \in T\}$ and $a_i = a_{S_i}$. Then $M \models \varphi(a_i, b_T)$ if and only if $T \in S_i$, that is, if and only if $i \in T$. So for every $T \subseteq \{1, \dots, n\}$ the tuple b_T satisfies $\varphi^{\text{opp}}(b_T, a_i)$ exactly for $i \in T$, and $a_1, \dots, a_n$ are shattered by φ^{opp} . Applying this in both directions gives the equivalence. 



Remark 5.1.6. A formula $\psi(x, y)$ has the *strict order property* if for every n some model M of T contains tuples $b_1, \dots, b_n$ with $\psi(M, b_1) \subsetneq \psi(M, b_2) \subsetneq \dots \subsetneq \psi(M, b_n)$, where $\psi(M, b) = \{a \in M : M \models \psi(a, b)\}$. The formula $x < y$ of Example 5.1.3 is the basic case. Shelah proved that a theory is unstable if and only if some formula has the independence property or some formula has the strict order property [57, Chapter II]. So an unstable theory either encodes arbitrary subsets, as the random graph does, or encodes an infinite order; NIP theories are those in which only the second reason can occur.

5.2. The real-valued case

As in Section 1.2, we now replace formulas by functions on arbitrary sets. On arbitrary sets compactness is not available, so shattering is required only for finite patterns.

Definition 5.2.1. Let A, B be sets and $\varphi : A \times B \rightarrow \mathbb{R}$ a bounded function. A sequence $(b_n)_{n < \omega}$ in B is *independent for φ* if there are reals $r < s$ such that for all disjoint finite sets $P, Q \subseteq \omega$ there is $a \in A$ with

$$\varphi(a, b_n) \geq s \text{ for } n \in P \quad \text{and} \quad \varphi(a, b_n) \leq r \text{ for } n \in Q.$$

φ has the *independence property* if some sequence in B is independent for φ , and φ is NIP otherwise.

For a $\{0, 1\}$ -valued function, any $0 < r < s < 1$ may be used, and a sequence is independent exactly when each finite part of the sequence is shattered in the sense of Definition 5.1.1. By compactness, a formula $\varphi(x, y)$ is NIP in the sense of Definition 5.1.1 if and only if the corresponding function on $M^{|x|} \times M^{|y|}$ is NIP for every model $M \models T$.

Stability was characterized in Section 1.2 by the exchange of iterated limits. NIP is characterized by the existence of convergent subsequences. For $b \in B$ let $e_b : \text{LS}(\varphi) \rightarrow [0, 1]$ be the evaluation map $e_b(p) = p(b)$, as in Subsection 1.2.4. The analytic input is Rosenthal's dichotomy.

Fact 5.2.2 (Rosenthal). *Let X be a set and (f_n) a uniformly bounded sequence of real-valued functions on X . Then (f_n) has a subsequence that converges pointwise on X , or a subsequence (f_{n_k}) for which there are reals $r < s$ such that for all disjoint finite sets $P, Q \subseteq \omega$ there is $x \in X$ with $f_{n_k}(x) \geq s$ for $k \in P$ and $f_{n_k}(x) \leq r$ for $k \in Q$.*

Theorem 5.2.3. *For $\varphi: A \times B \rightarrow [0, 1]$ the following are equivalent.*

- (1) φ is NIP.
- (2) For every sequence (b_n) in B , the sequence of functions (e_{b_n}) has a subsequence that converges pointwise on $\text{LS}(\varphi)$.

Proof. (2) $\Rightarrow$ (1). Suppose (b_n) is independent for φ , with reals $r < s$, and let (b_{n_k}) be a subsequence. For each $K < \omega$ choose $a_K \in A$ with $\varphi(a_K, b_{n_k}) \geq s$ for even $k < K$ and $\varphi(a_K, b_{n_k}) \leq r$ for odd $k < K$. Let $\mathcal{U}$ be a nonprincipal ultrafilter on ω and $p = \lim_{K, \mathcal{U}} \text{ltp}_\varphi(a_K)$, a point of $\text{LS}(\varphi)$. For each fixed k the inequality for k holds for all $K > k$, so $e_{b_{n_k}}(p) = p(b_{n_k}) \geq s$ for even k and $\leq r$ for odd k . Hence $(e_{b_{n_k}}(p))_k$ does not converge. Since the subsequence was arbitrary, (2) fails.

(1) $\Rightarrow$ (2). Suppose (2) fails for (b_n) . By Fact 5.2.2 applied to $X = \text{LS}(\varphi)$ and $f_n = e_{b_n}$, there are a subsequence (b_{n_k}) and reals $r < s$ such that for all disjoint finite $P, Q \subseteq \omega$ there is $p \in \text{LS}(\varphi)$ with $p(b_{n_k}) \geq s$ for $k \in P$ and $p(b_{n_k}) \leq r$ for $k \in Q$. Fix $r < r' < s' < s$. Since p is a pointwise limit of left types $\text{ltp}_\varphi(a)$ and only finitely many coordinates are involved, some $a \in A$ satisfies $\varphi(a, b_{n_k}) > s'$ for $k \in P$ and $\varphi(a, b_{n_k}) < r'$ for $k \in Q$. So (b_{n_k}) is independent for φ , with $r' < s'$, and φ is not NIP. 

The proof of (2) $\Rightarrow$ (1) above is elementary; only (1) $\Rightarrow$ (2) needs Rosenthal's dichotomy. The next proposition shows directly, using only the separately continuous extension of Proposition 1.2.8, that stability implies NIP also in the real-valued setting.

Proposition 5.2.4. *Every stable function $\varphi: A \times B \rightarrow [0, 1]$ is NIP.*

Proof. Let $\hat{\varphi}: \text{LS}(\varphi) \times \text{RS}(\varphi) \rightarrow [0, 1]$ be the separately continuous function of Proposition 1.2.8. Taking a constant family in (1.2) gives $\hat{\varphi}(p, \text{rtp}_\varphi(b)) = p(b)$ for all $p \in \text{LS}(\varphi)$ and $b \in B$.

Suppose, towards a contradiction, that (b_n) is independent for φ , with reals $r < s$. For $m < N$ choose $a_{m,N} \in A$ with

$$\varphi(a_{m,N}, b_n) \geq s \text{ for } m \leq n < N \quad \text{and} \quad \varphi(a_{m,N}, b_n) \leq r \text{ for } n < m,$$

and let $a_{m,N}$ be arbitrary for $N \leq m$. Fix a nonprincipal ultrafilter $\mathcal{U}$ on ω and put

$$p_m = \lim_{N, \mathcal{U}} \text{ltp}_\varphi(a_{m,N}) \in \text{LS}(\varphi).$$

For fixed n the displayed inequalities hold for all $N > \max(m, n)$, so

$$(5.1) \quad p_m(b_n) \geq s \text{ if } n \geq m, \quad p_m(b_n) \leq r \text{ if } n < m.$$

Now put $q = \lim_{n, \mathcal{U}} \text{rtp}_\varphi(b_n) \in \text{RS}(\varphi)$ and $p = \lim_{m, \mathcal{U}} p_m \in \text{LS}(\varphi)$. We compute $\hat{\varphi}(p, q)$ in two ways.

First, by continuity of $\hat{\varphi}(p_m, \cdot)$ and (5.1), $\hat{\varphi}(p_m, q) = \lim_{n, \mathcal{U}} p_m(b_n) \geq s$ for every m , since $p_m(b_n) \geq s$ for all $n \geq m$. By continuity of $\hat{\varphi}(\cdot, q)$, $\hat{\varphi}(p, q) = \lim_{m, \mathcal{U}} \hat{\varphi}(p_m, q) \geq s$.

Second, since limits in $\text{LS}(\varphi)$ are computed coordinatewise, $p(b_n) = \lim_{m, \mathcal{U}} p_m(b_n) \leq r$ for every n , because $p_m(b_n) \leq r$ for all $m > n$. By continuity of $\hat{\varphi}(p, \cdot)$, $\hat{\varphi}(p, q) = \lim_{n, \mathcal{U}} \hat{\varphi}(p, \text{rtp}_\varphi(b_n)) = \lim_{n, \mathcal{U}} p(b_n) \leq r$.

So $s \leq \hat{\varphi}(p, q) \leq r$, contradicting $r < s$. 

The types p_m in the proof of Proposition 5.2.4 form a ladder in $\text{LS}(\varphi)$ against the b_n , which is exactly what the separate continuity of $\hat{\varphi}$ forbids: an independent sequence is shattered in every pattern, and a staircase is one of the patterns.



Remark 5.2.5. Proposition 5.2.4 can also be derived, with heavier machinery, from Theorem 5.2.3. By Grothendieck’s double limit criterion, in the form of Proposition A.1.6, stability of φ makes $\{e_b : b \in B\}$ relatively weakly compact in $C(\text{LS}(\varphi))$. By Corollary A.1.5, which rests on the Eberlein–Šmulian theorem (see Fact A.1.4 below), every sequence (e_{b_n}) then has a pointwise convergent subsequence. Thus condition (2) of Theorem 5.2.3 holds, and φ is NIP by the implication (2) $\Rightarrow$ (1), which does not use Rosenthal’s dichotomy. The direct proof above avoids both Grothendieck’s criterion and the Eberlein–Šmulian theorem.



Remark 5.2.6. Theorem 1.2.9 represents stable functions by the pairing of a reflexive Banach space E with E^* . Khanaki [5] proved that φ is stable, NIP or NSOP according as a Banach space naturally associated with φ is reflexive, Rosenthal (contains no copy of ℓ_1) or weakly sequentially complete.

5.3. Historical remarks



Section 5.1. The independence property was introduced by Shelah [36], in the same paper as the order property. His theorem that an unstable theory has the independence property or the strict order property is in [57, Chapter II]. The combinatorial notion of shattering, and the VC dimension, were introduced independently in probability and learning theory by Vapnik and Chervonenkis [15]; the polynomial bound on the number of traces of a family of finite VC dimension was found independently by Sauer [11] and by Shelah [12]. Laskowski [6] observed that a formula is NIP exactly when the family of sets defined by the formula has finite VC dimension, connecting model theory with learning theory.



Section 5.2. Rosenthal’s dichotomy comes from his characterization of Banach spaces containing ℓ_1 [9], and the systematic study of pointwise compact sets of Baire class 1 functions is due to Rosenthal [10] and to Bourgain, Fremlin and Talagrand [1]. Their work entered model theory much later. The development is summarized in the following timeline.



1971

Shelah; Vapnik–Chervonenkis. Shelah [36] introduces the independence property. Independently, Vapnik and Chervonenkis [15] introduce shattering and what is now called VC dimension, in the study of uniform convergence of empirical frequencies.

1972

Sauer; Shelah. Sauer [11] and Shelah [12] independently prove the polynomial bound on the number of traces of a set system of finite VC dimension.

- 1974–1978

Rosenthal; Bourgain–Fremlin–Talagrand. Rosenthal proves the ℓ_1 dichotomy [9] and studies pointwise compact sets of Baire class 1 functions [10]. Bourgain, Fremlin and Talagrand [1] develop the theory of these “Rosenthal compacta”.
- 1981

Poizat. Poizat [8] characterizes countable NIP theories by a bound on the number of types, and develops the theory of unstable theories without the independence property.
- 1986

Pillay–Steinhorn. Pillay and Steinhorn [7] develop the theory of o-minimal structures, following van den Dries. These structures are NIP and are the main source of unstable NIP theories.
- 1992

Laskowski. Laskowski [6] identifies NIP formulas with formulas defining families of finite VC dimension.
- 2008–2011

Hrushovski–Peterzil–Pillay; Hrushovski–Pillay. The study of invariant and generically stable measures in NIP theories [3, 4], motivated by Pillay’s conjecture on definable groups in o-minimal structures, turns NIP into a central topic of model theory.
- 2009

Ben Yaacov. Ben Yaacov [2] extends VC classes to $[0, 1]$ -valued functions, the continuous analogue of NIP used in Section 5.2.
- 2015

Simon. Simon [14] applies the theory of Rosenthal compacta to spaces of φ -types of NIP formulas, and publishes the monograph [13] on NIP theories.
- 2015–2020

Khanaki. Khanaki [20, 5] relates stability, NIP and NSOP to reflexive, Rosenthal and weakly sequentially complete Banach spaces, respectively.

APPENDIX A

Grothendieck's double limit criterion

Grothendieck's double limit criterion is used at several points in these notes: in Subsection 1.2.4 and in Remarks 1.2.14 and 5.2.5. This appendix states the criterion in the form we need and explains how the criterion relates to the stability of functions (see Definition 1.2.5). Throughout, K is a compact Hausdorff space and $C(K)$ is the Banach space of continuous real-valued functions on K with the supremum norm. The *weak topology* on $C(K)$ is the coarsest topology making every bounded linear functional continuous, and the *pointwise topology* is the one induced by the product topology of $\mathbb{R}^K$. Point evaluations $f \mapsto f(x)$ are bounded linear functionals, so the weak topology is finer than the pointwise topology.

A.1. The criterion

Definition A.1.1. A bounded set $H \subseteq C(K)$ satisfies the *double limit condition* if for all sequences (x_m) in K and (f_n) in H and all ultrafilters $\mathcal{U}, \mathcal{V}$ on $\mathbb{N}$,

$$\lim_{m, \mathcal{U}} \lim_{n, \mathcal{V}} f_n(x_m) = \lim_{n, \mathcal{V}} \lim_{m, \mathcal{U}} f_n(x_m).$$

In the language of Definition 1.2.5, this says that the evaluation function $K \times H \rightarrow \mathbb{R}$, $(x, f) \mapsto f(x)$, is stable.

Grothendieck stated the condition with ordinary limits. The two formulations are equivalent.

Lemma A.1.2. *A bounded set $H \subseteq C(K)$ satisfies the double limit condition if and only if*

$$\lim_m \lim_n f_n(x_m) = \lim_n \lim_m f_n(x_m)$$

for all sequences (x_m) in K and (f_n) in H for which both iterated limits exist.

Proof. If both ordinary iterated limits exist, these limits are equal to the iterated limits along any nonprincipal ultrafilters, so the double limit condition implies the second condition.

Conversely, suppose the double limit condition fails for (x_m) , (f_n) , $\mathcal{U}$ and $\mathcal{V}$, with iterated limits $\alpha \neq \beta$. After an affine change of scale we may assume that all values lie in $[0, 1]$. Proposition 1.2.3, applied to the function $(x, f) \mapsto f(x)$, gives subsequences, which we still denote (x_m) and (f_n) , with $|f_n(x_m) - \alpha| \leq 2^{1-m}$ for $m < n$ and $|f_n(x_m) - \beta| \leq 2^{1-n}$ for $n < m$. By a diagonal argument we may pass to further subsequences along which $\lim_n f_n(x_m)$ exists for every m and $\lim_m f_n(x_m)$ exists for every n ; the inequalities are preserved, with the indices renumbered. Then $\lim_n f_n(x_m)$ is within 2^{1-m} of α , so $\lim_m \lim_n f_n(x_m) = \alpha$, and symmetrically $\lim_n \lim_m f_n(x_m) = \beta$. So the second condition fails. ☺

Theorem A.1.3 (Grothendieck). *Let $H \subseteq C(K)$ be bounded. The following are equivalent.*

- (1) *H is relatively weakly compact in $C(K)$.*
- (2) *Every function in the pointwise closure of H in $\mathbb{R}^K$ is continuous.*
- (3) *H satisfies the double limit condition.*

Proof. (2) $\Rightarrow$ (3). Let (x_m) , (f_n) , $\mathcal{U}$, $\mathcal{V}$ be given, and put $x = \lim_{m,\mathcal{U}} x_m \in K$ and $f = \lim_{n,\mathcal{V}} f_n \in \mathbb{R}^K$. By (2), f is continuous. Using the continuity of f and of each f_n ,

$$\lim_{m,\mathcal{U}} \lim_{n,\mathcal{V}} f_n(x_m) = \lim_{m,\mathcal{U}} f(x_m) = f(x) = \lim_{n,\mathcal{V}} f_n(x) = \lim_{n,\mathcal{V}} \lim_{m,\mathcal{U}} f_n(x_m).$$

(3) $\Rightarrow$ (2). Suppose f is in the pointwise closure of H but is not continuous at some $x \in K$. Then $f = \lim_{i,\mathcal{U}} f_i$ for a family $(f_i)_{i \in I}$ in H and an ultrafilter $\mathcal{U}$ on I , and there are a family $(x_j)_{j \in J}$ in K and an ultrafilter $\mathcal{V}$ on J with $\lim_{j,\mathcal{V}} x_j = x$ and $\lim_{j,\mathcal{V}} f(x_j) = \beta \neq f(x)$. (Fix $\varepsilon > 0$ such that every neighborhood O of x contains a point x_O with $|f(x_O) - f(x)| \geq \varepsilon$, and let $\mathcal{V}$ be an ultrafilter on the set of neighborhoods of x containing, for each O , the set of neighborhoods contained in O .) By continuity of each f_i ,

$$\lim_{i,\mathcal{U}} \lim_{j,\mathcal{V}} f_i(x_j) = \lim_{i,\mathcal{U}} f_i(x) = f(x), \quad \lim_{j,\mathcal{V}} \lim_{i,\mathcal{U}} f_i(x_j) = \lim_{j,\mathcal{V}} f(x_j) = \beta.$$

After an affine change of scale, Proposition 1.2.3 and the proof of Corollary 1.2.6 turn these families into sequences whose iterated limits along a nonprincipal ultrafilter are β and $f(x)$. This contradicts (3).

(1) $\Rightarrow$ (2). Let W be the weak closure of H , which is weakly compact by (1). The identity map from $C(K)$ with the weak topology to $C(K)$ with the pointwise topology is continuous, so W is pointwise compact, hence closed in $\mathbb{R}^K$. So the pointwise closure of H is contained in $W \subseteq C(K)$.

(2) $\Rightarrow$ (1). Let $\bar{H}$ be the pointwise closure of H . This set is compact in $\mathbb{R}^K$, because H is bounded, and is contained in $C(K)$ by (2). We need only show that the weak and pointwise topologies agree on $\bar{H}$, since then $\bar{H}$ is weakly compact. By the Riesz representation theorem, the weak topology is generated by the maps $h \mapsto \int h d\mu$ with μ a Radon measure on K . On the uniformly bounded set $\bar{H}$ each such map is sequentially continuous for the pointwise topology, by Lebesgue's dominated convergence theorem. Grothendieck proved that a pointwise compact subset of $C(K)$ is *angelic*: every point in the closure of a subset is the limit of a sequence from that subset [44]; see also Pryce [25]. On an angelic space sequentially continuous functions are continuous, which completes the proof. $\odot$

The proof of (2) $\Leftrightarrow$ (3) in Theorem A.1.3 uses only ultrafilter limits and Proposition 1.2.3. The measure-theoretic input and the angelicity of pointwise compact sets enter only in (2) $\Rightarrow$ (1).

Fact A.1.4 (Eberlein–Šmulian). *In a Banach space, every sequence in a relatively weakly compact set has a weakly convergent subsequence [16, 17].*

Corollary A.1.5. *If a bounded set $H \subseteq C(K)$ satisfies the double limit condition, then every sequence in H has a subsequence that converges pointwise on K , to a continuous function.*

Proof. By Theorem A.1.3, H is relatively weakly compact. By Fact A.1.4, every sequence in H has a subsequence converging weakly to some $f \in C(K)$. Point evaluations are bounded linear functionals, so the subsequence converges pointwise to f . $\odot$

We now connect the criterion with the spaces of types of Section 1.2. For $\varphi: A \times B \rightarrow [0, 1]$, recall the evaluation maps $e_b: \text{LS}(\varphi) \rightarrow [0, 1]$, $e_b(p) = p(b)$, and $e_a: \text{RS}(\varphi) \rightarrow [0, 1]$, $e_a(q) = q(a)$, of Subsection 1.2.4.

Proposition A.1.6. *Let $\varphi: A \times B \rightarrow [0, 1]$ be stable. Then $\{e_b : b \in B\}$ is relatively weakly compact in $C(\text{LS}(\varphi))$, and $\{e_a : a \in A\}$ is relatively weakly compact in $C(\text{RS}(\varphi))$. More precisely, the pointwise closure of $\{e_b : b \in B\}$ consists of the functions $\hat{\varphi}(\cdot, q)$ with $q \in \text{RS}(\varphi)$, where $\hat{\varphi}$ is the function of Proposition 1.2.8.*

Proof. By (1.2) with a constant family, $e_b = \hat{\varphi}(\cdot, \text{rtp}_\varphi(b))$. If $q = \lim_{j, \nu} \text{rtp}_\varphi(b_j)$, then by continuity of $\hat{\varphi}(p, \cdot)$ we get $\lim_{j, \nu} e_{b_j}(p) = \hat{\varphi}(p, q)$ for every p . So every pointwise limit of functions e_b is of the form $\hat{\varphi}(\cdot, q)$, and conversely every such function is a pointwise limit of this kind. These functions are continuous on $\text{LS}(\varphi)$ by separate continuity. Hence condition (2) of Theorem A.1.3 holds, and $\{e_b : b \in B\}$ is relatively weakly compact. The statement for $\{e_a : a \in A\}$ is proved symmetrically. 

A.2. Historical remarks

 **Section A.1.** Theorem A.1.3 is from Grothendieck's 1952 paper [44], announced in [18], which also proves that pointwise compact subsets of $C(K)$ are angelic. The Eberlein–Šmulian theorem is due to Šmulian [16] and Eberlein [17]. Later proofs and extensions of the criterion, by Namioka, Pták, Young, Simons, Pryce and König, are listed in the timeline of Section 1.3 (see the historical remarks on Subsection 1.2.1). The reading of the criterion as a statement about stable functions, which is the point of view of Definition A.1.1 and Proposition A.1.6, is due to Ben Yaacov [42] and Pillay [21] for formulas, following Iovino [49] and Krivine–Maurey [51].

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